

DATA SCIENCE WITH EXCEL

Exploratory Data Analysis for Banking Analysts

From raw tables to reliable insight — using the tools already available in Excel.

STRUCTURED DATA

EDA

STATISTICS

DATA QUALITY

Banking context · Excel-first examples · Practical formulas and charts

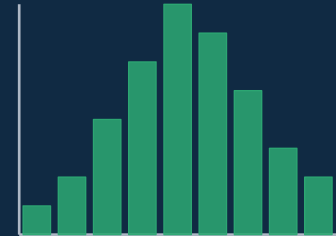
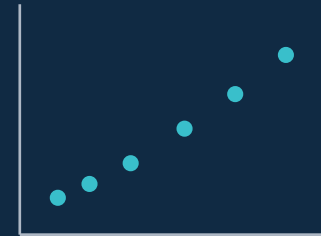
X

x Retail Banking EDA

fx =CORREL (Table1[Income],Table1[Balance])

ID	Age	Income	Balance
C1001	42	65k	12.5k
C1002	35	54k	8.2k
C1003	61	92k	45.5k
C1004	29		3.2k
C1005	47	71k	165k

Correlation



Course schedule across time zones

Comparing the Italy schedule with Eastern Europe participants (Romania and Hungary)



SAME
LEARNING
GREATER
TOGETHER



Current schedule



Italy time



Morning session
09:30 – 13:00



Afternoon session
14:00 – 17:30



+1 hour



Eastern Europe time

Romania & Hungary



Morning session
10:30 – 14:00



Afternoon session
15:00 – 18:30



Observation

The current Italy schedule becomes quite late for Eastern Europe, especially in the afternoon.



Recommended adjustment



Proposal: start one hour earlier



This keeps the course more comfortable for Eastern Europe participants.



Proposed Italy time



Morning session
08:30 – 12:00



Afternoon session
13:00 – 16:30



+1 hour



Eastern Europe time

Romania & Hungary



Morning session
09:30 – 13:00



Afternoon session
14:00 – 17:30



Best practice: when training an international audience, always check local time equivalents before confirming the timetable.

Course Schedule

Dual timetable for Italy / Hungary and Romania



*Same learning opportunities.
Different time zones.
Shared goals.*



Italy / Hungary

SAME COURSE
SHARED PROGRESS

	09:00 – 10:30	Session
	10:30 – 10:40	Morning break (10 min)
	10:40 – 12:00	Session
	12:00 – 13:00	Lunch break
	13:00 – 15:00	Session
	15:00 – 15:15	Afternoon break (15 min)
	15:15 – 17:00	Session



Daily timetable: 09:00 – 12:00 / 13:00 – 17:00



Romania

SAME CONTENT
LOCAL CONVENIENCE

	10:00 – 11:30	Session
	11:30 – 11:40	Morning break (10 min)
	11:40 – 13:00	Session
	13:00 – 14:00	Lunch break
	14:00 – 16:00	Session
	16:00 – 16:15	Afternoon break (15 min)
	16:15 – 18:00	Session

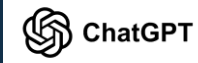


Daily timetable: 10:00 – 13:00 / 14:00 – 18:00



Why this structure?

The **morning block** is slightly shorter and the afternoon block longer. This is intentional: it helps avoid pushing the lunch break too late in Romania. In exchange, the mid-morning break is only 10 minutes, while the afternoon break is 15 minutes.



The first draft of this set of slides was generated using AI tools under the direction of Antonio Piemontese and in accordance with his general and specific instructions and guidelines. The material was subsequently reviewed, revised, and further refined through additional interactions with AI tools, based on Antonio Piemontese's feedback and specific requests for changes. Throughout the process, Antonio Piemontese retained responsibility for the overall direction, content, review, and final approval of the material.

This declaration is provided in accordance with the transparency requirements of the European Union's AI Act.

Course map: what we will build together

A conceptual framework for data analysis, anchored in Excel and banking examples.

0	Preliminaries	Technical Requirements, Language, Anatomy of Excel	
1	Foundations	Data Science, analytics, banking context	
2	Data & Types	Structured tables and data types	
3	EDA	Profiling, summaries, pivots, visual exploration	
4	Probability & Statistics	Random variables, distributions, uncertainty	
5	Relationships	Correlation, correlation matrices and patterns	
6	Visualisation	Plots that reveal structure and problems, dashboards	
7	Data issues	Outliers, missing values, quality	

Goal: make the analysis explainable, auditable and reproducible — not just “get a chart”.

Chapter title



00 - Preliminaries

Technical requirements: Excel version matters

Minimum setup to run every example in this training deck.

- 1 **Microsoft Excel 2021 or later — minimum for full compatibility.**
- 2 **Recommended: Microsoft Excel 2024 or Microsoft 365.**
- 3 Enable the Analysis ToolPak (Data → Data Analysis).
- 4 Windows 10 or Windows 11 recommended for the classroom setup.
- 5 PowerPoint 2016+ or a compatible presentation viewer.
- 6 Dynamic-array functions are used in examples: SORT and UNIQUE.
- 7 Excel Tables (Ctrl+T) are used throughout the EDA workflow.
- 8 No Python, R, Power BI or other Data Science software required.
- 9 No VBA or macros required.
- 10 No Internet connection required for the exercises.

KEY REQUIREMENT Full compatibility requires Excel 2021 or later • Recommended: Excel 2024 / Microsoft 365

How to change the language in Excel

Display language controls menus and tabs; authoring language affects editing and localized function names.

1 Open Options

File → Options → Language (with possible download for the new language)

2 Choose the language

Office display language

Changes Ribbon tabs, menus, commands and dialog boxes.

Office authoring languages

Controls proofing/editing language and can affect localized function names.

Select the desired language → Set as Preferred / Default → restart Excel if requested.

What changes?

Ribbon

Home ↔ Home
Data ↔ Dati

Menus

File ↔ File
Options ↔ Opzioni

Functions

SUM ↔ SOMMA
AVERAGE ↔ MEDIA

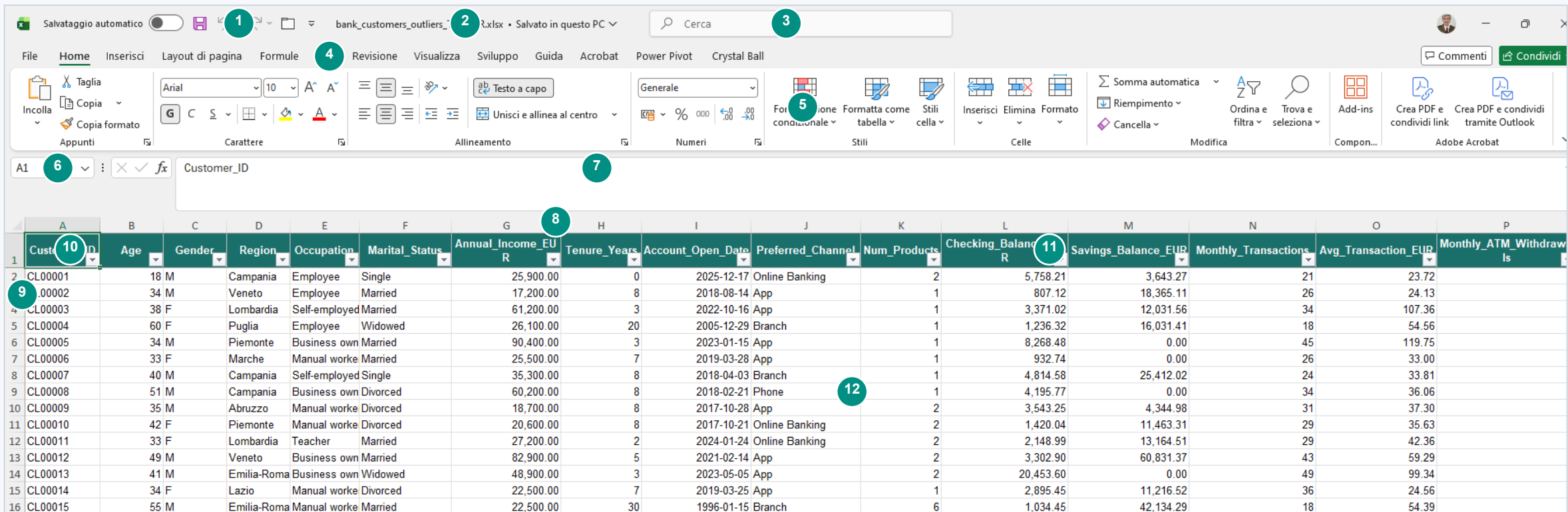
Separators

Comma / semicolon and decimal conventions may depend on regional settings.

If the language is not installed, Excel may offer Add a Language / install the Office language pack.

Anatomy of the Excel interface (2024)

The main components you will use when exploring and analysing tabular data



1 Quick Access Toolbar

Frequently used commands such as Save, Undo and Redo.

2 Title bar

Shows the workbook name and the Excel application window.

3 Search box

Find commands, help topics and Excel features quickly.

4 Ribbon tabs

Home, Insert, Formulas, Data, Review, View, etc.

5 Ribbon

Context-sensitive command area organised into groups.

6 Name Box

Shows the active cell reference; can also jump to a cell/range.

7 Formula Bar

View, enter and edit values or formulas for the active cell.

8 Column headings

Letters identify columns; click to select an entire column.

9 Row headings

Numbers identify rows; click to select an entire row.

10 Active cell

The currently selected cell; its reference appears in the Name Box.

11 Table header & filters

Field names plus sorting/filter controls in an Excel Table.

12 Worksheet grid

The cell area where tabular data, formulas and results live.

Also important (not visible in this screenshot): sheet tabs, horizontal/vertical scroll bars, the status bar and zoom controls.

Excel Formulas in Italian and English

The main matchings

English Excel	Italian Excel	Where we use it
IF	SE	Checks: every flag
AND / OR	E / O	Checks: mortgage vs payment, age, date, score
COUNT	CONTA.NUMERI	Statistics: N numeric values
COUNTA	CONTA.VALORI	Key: total injected anomalies
COUNTBLANK	CONTA.VUOTE	Statistics: blank cells; sentinels and blanks
COUNTIF	CONTA.SE	Statistics: values beyond the fences; Checks: duplicate ID, summary
COUNTIFS	CONTA.PIÙ.SE	Checks: duplicate record; cluster (48 × 199)
AVERAGE	MEDIA	Statistics: mean; z-score
MEDIAN	MEDIANA	Statistics: median
STDEV	DEV.ST	Statistics: standard deviation; z-score
MIN / MAX	MIN / MAX	Statistics; range checks
QUARTILE	QUARTILE	Statistics: Q1, Q3 → IQR
SUMPRODUCT	MATR.SOMMA.PRODOTTO	Checks: N flags and summary
ABS	ASS	Checks: date vs tenure
INT	INT	Checks: date vs tenure

VLOOKUP	CERCA.VERT	Checks: look up a customer by ID in another sheet (e.g. Key)
HLOOKUP	CERCA.ORIZZ	Checks: look up a value by column header in a horizontal table
DATE	DATA	Checks: extraction date DATE(2026,9,13)
YEAR	ANNO	Checks: plausible date
TODAY	OGGI	Key: filtering future dates
DATEDIF	DATA.DIFF	Key: date/tenure consistency
LEN	LUNGHEZZA	Key: trailing spaces in categories
TRIM	ANNULLA.SPAZI	Key: trailing spaces in categories
UPPER	MAIUSC	Key: standardising categories

Three modern variants participants may type instead of the "classic" functions used in the file: STDEV.S → DEV.ST.C, QUARTILE.INC → INC.QUARTILE, STANDARDIZE → NORMALIZZA (computes the z-score directly).

The same formula in both languages, separators included — English Excel uses `,` between arguments and `.` for decimals, Italian Excel uses `;` and `,`:

`=IF(OR(B2<18,B2>100),"Implausible age","")` ↔ `=SE(O(B2<18;B2>100);"Implausible age";"")` and `=Q1-1.5*IQR` ↔ `=Q1-1,5*IQR`.

Two main behaviours when editing a formula

Especially regarding the arrow keys

When you are editing a formula in Excel, there are **two main behaviors**, especially for the arrow keys:

- **Edit mode:** the ← → ↑ ↓ keys move the cursor inside the formula, without changing the cell references.
- **Point mode:** the arrow keys select/move the reference to a cell in the formula.

The F2 key allows you to switch to the formula text editing behavior.

For example, with:

=A1+\$B\$2

if the cursor is on A1 :

- in **Edit mode**, pressing → moves the cursor between the characters;
- in **Point mode**, pressing → can change the selected reference from A1 to B1.

POINT MODE:
FORMULA SELECTION

EDIT MODE:
IN-CELL EDITING

CELL MODES

READY

ENTER

POINT

EDIT

(Pronto, Invio, Puntatore, Modifica)

Knowing Excel cell modes (Ready, Enter, Point, and Edit) helps you [avoid frustrating navigation errors and fix broken formula references quickly!](#)

The default Edit mode

Arrow keys move the cursor inside the formula.

Ready  =A1|+\$B\$2



Moves the cursor within the formula text.

Point mode

Arrow keys select/move the reference to a cell.

fx =B1+\$B\$2

	A	B	C
1	A1	B1	
2			
3			



Changes the selected reference (from A1 to B1).

Ready

click on right mouse button in the left bottom corner (and then on the top line) to see cell modes

fx =A1+A2

Editing
formulas

 Name
Manager

Formulas

 Conditional
Formatting

Home

 Data
Validation

Data

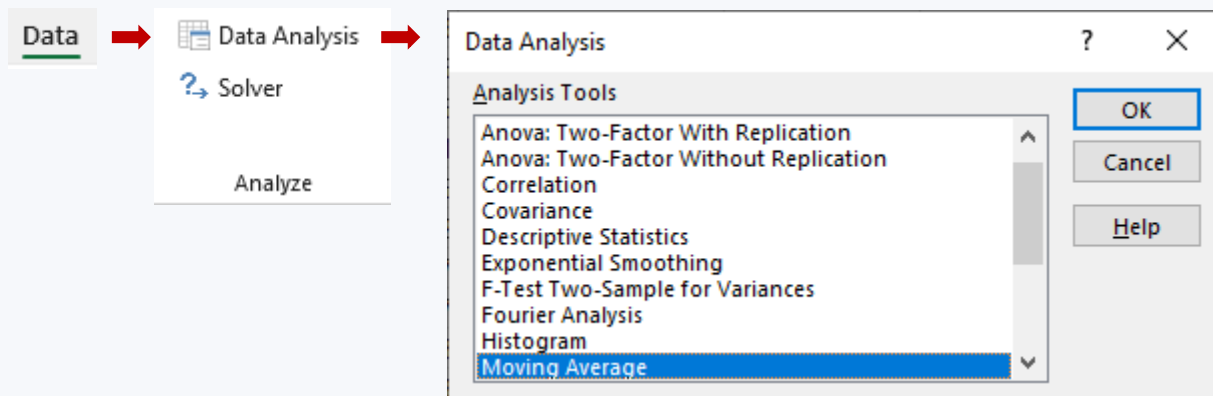
 Select Data
Chart Design

Insert

link to a good
explanation
video

Analysis tool –a key piece of functionality

What it is (our target)



How to get it installed

The screenshot shows the 'Excel Options' dialog box with the 'Add-ins' tab selected. The 'Add-ins' list shows the 'Analysis ToolPak' as an inactive add-in. The steps for installation are numbered 1 through 5:

1. Click on the 'File' menu.
2. Click on the 'Options' button.
3. Click on the 'Add-ins' tab in the 'Excel Options' dialog box.
4. Click on the 'Go...' button next to the 'Excel Add-ins' dropdown.
5. In the 'Add-ins available' dialog box, check the 'Analysis ToolPak' checkbox and click 'OK'.

Cell References

In Excel, **A1 references** are the standard way to identify cells using:

- a **column letter**
- followed by a **row number**

Examples:

```
A1  
B5  
M22  
AA100
```

So:

- **A1** = column A, row 1
- **M22** = column M, row 22

This is called **A1 reference style**.

It is also used for ranges:

```
A1:A10  
B2:D20  
M22:M31
```

Excel also distinguishes:

- **relative reference** → **A1**
- **absolute reference** → **\$A\$1**
- **mixed reference** → **\$A1** or **A\$1**

Example:

```
</> excel  
  
=A1*$B$1
```

If copied down, **A1** changes, while **\$B\$1** stays fixed.

The alternative is the **R1C1 reference style**, for example:

```
R1C1  
R22C13
```

where **R** means row and **C** means column.

For most users and for the training deck, **A1 notation is the normal/default Excel notation.**

Chapter title



01 - Foundations

Data Science in one slide

Data Science combines business questions, data, statistical reasoning and computational tools to support decisions.



What matters most

A model or dashboard is useful only if it answers a relevant question with trustworthy data.

Where Excel fits

Excel is excellent for inspection, quick EDA, small-to-medium datasets, validation and communication.

Where Excel stops

Very large datasets, heavy automation and advanced modelling often need SQL, Python, R or cloud platforms.

Question → Data → Evidence → Decision

Why Excel still matters for bank analysts

Excel is not “less scientific”; it is a practical analysis surface when used with disciplined methods.

Fast inspection

Open a table and immediately inspect values, types, gaps and extremes.

Business proximity

Analysts, risk teams and finance users already exchange workbooks.

Transparent logic

Formulas, filters and PivotTables make many transformations visible.

Excellent EDA

Descriptive statistics, histograms, scatter plots, box plots and correlations.

Audit support

Checks, reconciliation and exception lists can be made explicit.

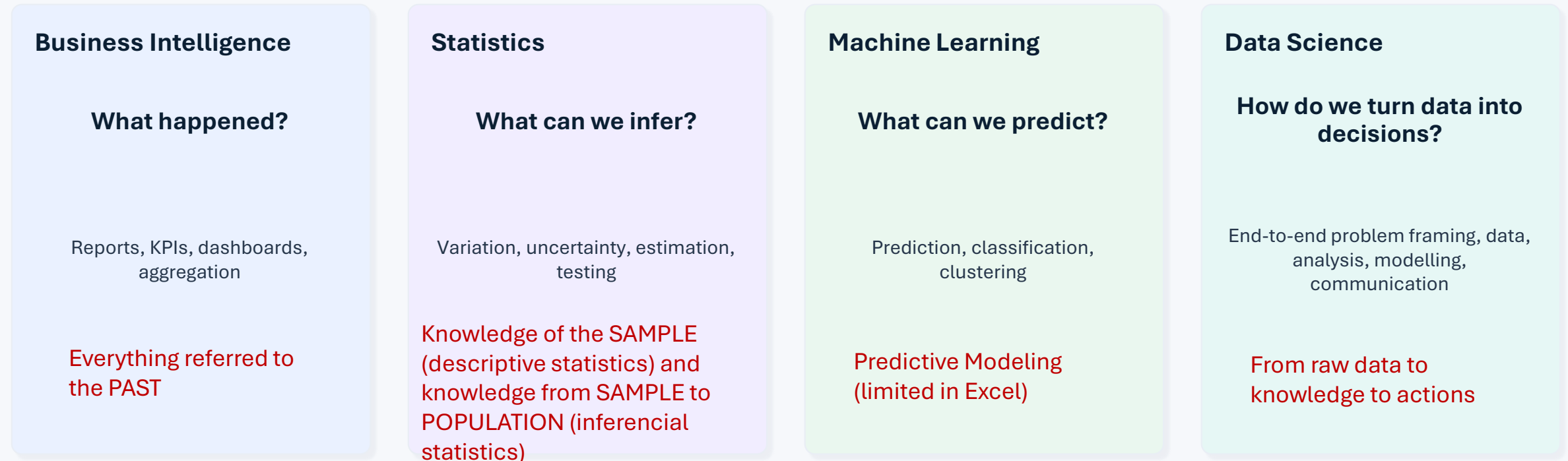
Bridge tool

Useful before/after SQL, Python or modelling pipelines.

The key limitation to Data Science is not the tool: it is uncontrolled data handling, hidden assumptions and poor reproducibility.

Data Science, statistics, BI and Machine Learning

They overlap, but they answer different kinds of questions.



Excel can support all four — but EDA and descriptive statistics are where it is especially strong.

Banking examples: questions before tools

Start with the decision or risk question, then choose the analysis.

Domain	Business question	Excel-first EDA
Retail customers	Which segments show declining balances?	PivotTable + line chart
Credit risk	Which variables move with arrears (ie, payments that are overdue or unpaid when due) or default?	Correlation + distributions
Operations	Where are processing times unusually high?	Histogram + outlier analysis
Fraud monitoring	Which transactions are atypical?	Patterns + anomaly flags
Data governance	Where is information incomplete or inconsistent?	Missingness + quality checks

The data landscape in a bank

Most analysis starts from structured operational data — but meaning comes from context.

Core banking

accounts · balances · products

Payments

transactions · merchants · channels

CRM

customers · campaigns · interactions

Risk systems

exposure · ratings · arrears

External data

rates · macro · geography

ETL processing
(Extract / Transform / Load)

Source, operational data

X

Analysis-ready extract

fx

=ROWS (Table1)

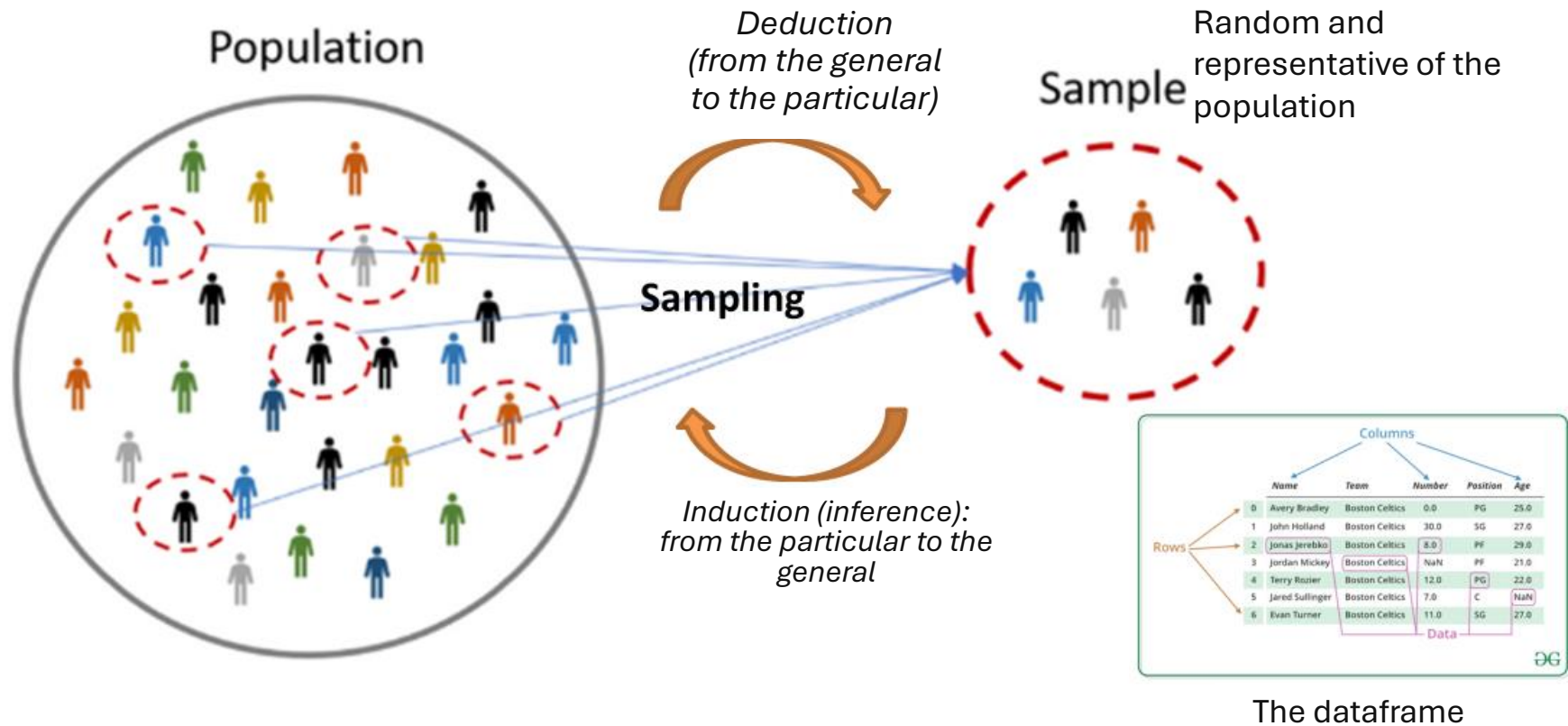
CustomerID	Date	Balance	Segment	Default
C1001	2026-08-31	12,500	Affluent	No
C1002	2026-08-31	8,200	Mass	No
C1003	2026-08-31	45,500	Affluent	No
C1004	2026-08-31	3,200	Mass	No
C1005	2026-08-31	165,000	Affluent	Yes
C1006	2026-08-31	9,700	Mass	No

A structured table ready to be analysed

Data Science with Excel · Banking Analysts

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Sample vs. Population



Excel 2024 limit per single worksheet is 1,048,576 rows.
Computing EDA and building predictive models over the full
population might be too demanding for our hardware.

Sampling selects observations from the population; statistical inference uses the sample to learn about the population.

Building a dataset in Excel: two starting points

A dataset can be created directly in Excel or imported from an external source. In both cases, the goal is a clean, structured, analysis-ready table.

1 BUILD THE DATASET NATIVELY IN EXCEL

Typical sources

- Manual data entry
- Copy / paste from controlled sources
- Form-based or operational data entry

Customer_ID	Age	Region	Balance
CL001	34	North	5,240
CL002	41	West	8,120
CL003	29	South	3,410

Good practice

- One row = one observation
- One column = one variable
- Use clear headers and consistent data types
- Convert the range to an Excel Table (Ctrl+T)

OR

2 IMPORT THE DATASET FROM OUTSIDE EXCEL

Typical external sources

- CSV / TXT files
- Other Excel workbooks
- Databases
- Web / API sources via Power Query

CSV / DB

Data →
Get Data

Review import
settings

Check before loading

- Delimiter and text encoding
- Locale (decimal and date conventions)
- Inferred data types
- Missing values and unexpected columns

ANALYSIS-READY
EXCEL TABLE

Key point: regardless of the starting point, verify structure, data types, missing values and business meaning before starting EDA.

Chapter title



02 – Data & Types

Structured vs semi-structured vs unstructured data

Data Science works with many kinds of data. Excel is strongest with structured data, but analysts often receive all three forms.

STRUCTURED DATA

tabular data

Clear rows, columns and fields

Examples: Excel tables, CSV files, relational databases, banking transaction tables
Typical fields: Customer_ID, Region, Balance, Date, Product_Count
Excel fit: excellent

Recognition: every row represents the same type of entity, and every column has a stable meaning.

SEMI-STRUCTURED DATA

Some organisation, but not fully tabular

Examples: JSON, XML, logs, email headers, API responses (there are also metadata)
Has labels or tags, but the shape may vary from record to record
Excel fit: possible after transformation or flattening

Recognition: records contain keys or tags, but not every record must contain the same fields.

UNSTRUCTURED DATA

No fixed table structure

Examples: PDFs, emails, contracts, images, audio, call recordings
Meaning exists, but structure is implicit rather than explicit
Excel fit: weak unless data is first extracted or summarised

Recognition: humans can understand the content, but software must work harder to extract variables.

Text, images, videos, sounds, speech, 3D

Excel takeaway: structured data is the natural starting point for EDA in Excel; semi-structured and unstructured data usually need preparation before they can be analysed as tables.

Structured data: the backbone of everyday Data Science

Structured data has an explicit schema: rows are observations, columns are variables, cells are values.

x Table1 · Customer snapshot				
fx =AVERAGE(Table1[Balance])				
CustomerID	Age	Income	Balance	Segment
C1001	42	65,000	12,500	Affluent
C1002	35	54,000	8,200	Mass
C1003	61	92,000	45,500	Affluent
C1004	29	—	3,200	Mass
C1005	47	71,000	165,000	Affluent

This in Data Science is TIDY DATA

Rows = observations

One customer, account, loan, transaction or time period per row.

Columns = variables

Age, balance, product, segment, status, date, risk score...

Schema = meaning

Names, types, units, valid ranges, keys and business definitions.

Anatomy of a table

Understanding the grain of the table prevents many analytical mistakes.

x Monthly customer snapshot			
fx =COUNTIFS(Table1[CustomerID],"C1001")			
CustomerID	Month	Balance	Products
C1001	Jan	11,800	3
C1001	Feb	12,100	3
C1001	Mar	12,500	3
C1002	Jan	8,700	2
C1002	Feb	8,450	2
C1002	Mar	8,200	2

Primary key / identifier

CustomerID + Month uniquely identifies each row.

Grain

One row = one customer in one month.

Risk of wrong grain

Averaging balances at the wrong level can double-count customers.

Observation, variable, value: three simple words, three different roles

A table becomes easier to reason about when these terms are explicit.

1

OBSERVATION (row)

A unit being analysed

Example: Customer C1003

2

VARIABLE (column)

A measured attribute

Example: AnnualIncome

3

VALUE (cell)

A concrete measurement

Example: €92,000

In Excel: rows usually represent observations, columns variables, and cells values.

What makes tabular data analysis-ready?

A clean table is a model of the business process, not merely a rectangular range.



One row, one observation

No merged cells; no subtotals inside the data area.



One column, one variable (a unique data type)

Do not mix currency, text and comments in the same field.



Consistent units

EUR vs thousands of EUR must be explicit.



Stable categories

“Retail”, “RETAIL”, “retail” should not be three categories.



Explicit missing values

Blank ≠ zero ≠ not applicable.



Traceable definitions

Every field has a business meaning and owner.

Tip: convert ranges to an Excel Table (Ctrl+T) before serious EDA.

Why convert a range to an Excel Table?

Before starting EDA, turn a normal cell range into a formal Excel Table (Ctrl+T).

Structured reference	Meaning
Table1[Col1]	Entire Col1
Table1[@Col1]	Col1 in the current row
[@Col1]	Same, when used inside Table1

NORMAL RANGE

Example: A1:F1000

ID	Age	Income	Bal.	Seg.
C01	42	65k	12.5k	A
C02	35	54k	8.2k	M
C03	61	92k	45.5k	A
C04	29	—	3.2k	M

Typical formula:
=AVERAGE(D2:D1000)



Ctrl+T

EXCEL TABLE

A structured object designed for analysis

- **Stable headers** Column names remain explicit and usable.
- **Built-in filters** Sort and filter every column immediately.
- **Automatic expansion** New rows and formulas are included automatically.
- **Structured references** Use readable formulas such as Table1[Balance].

```
=AVERAGE (Table1[Balance])
```

For EDA, an Excel Table makes the analysis more robust, readable and less error-prone.

The special behaviour of an Excel Table?

Here are the behaviours that set a table (Ctrl+T) apart from a plain range:

Structure and headers

- Headers must be text and unique: a number is converted to text, a duplicate name becomes "Name2".
- • When scrolling down with the active cell inside the table, the headers replace the column letters: no need to freeze panes.
- • Every header has a filter button (can be switched off in Table Design ► Filter Button); sorting and filtering always move entire rows.
- • Table cells cannot be merged; a table cannot overlap another table or a PivotTable.

Automatic expansion

- • Typing in the adjacent row or column extends the table on its own; Tab on the last cell creates a new row.
- • New rows inherit formats, data validation, conditional formatting and column formulas: this is why the 12 rules in the file follow the table when customers are added.
- Inserting or deleting rows/columns inside the table (Insert Table Rows Above, etc.) does not affect the cells outside the table on the same sheet row.
- • The handle in the bottom-right corner and Table Design ► Resize Table resize it manually.

Formulas

- Calculated columns: a formula entered in one cell fills the whole column and stays consistent; if a cell is edited by hand, Excel flags the exception.
- • Structured references: `tblCustomers[Balance]`, `[@Balance]` (current row), `[#Headers]`, `[#Totals]`, `[#All]`; they adjust automatically as the table grows, whereas `$L$2:$L$401` stays fixed. This is why the A1-based counts in the Rules sheet do not update when rows are added, while `tblCustomers[...]` would.
- • Important exception: structured references are not allowed in conditional-formatting rules (nor in data validation).
- Dynamic-array formulas cannot spill inside a table (#SPILL!).

Total row

- Table Design ► Total Row adds a row with a drop-down per column; the functions use `SUBTOTAL(109,...)`, which ignores filtered or hidden rows.

Formatting

- • Table styles with banded rows/columns and highlighted first/last column; changing the style updates everything, and direct formatting overrides the style.
- • Table Design ► Convert to Range removes the table while keeping its look; the style can be reset with Clear.

Integration

- • The table name is a ready-made dynamic range for PivotTables, charts, data validation and Power Query (Data ► From Table/Range): on refresh they pick up the new rows without changing the source.
- Slicers are available directly on the table (Table Design ► Insert Slicer), not only on PivotTables.
- • Remove Duplicates and the Table Design tab are always at hand.

Quick selection

- • Black arrow on the top edge of a header: one click selects the column data, two clicks include the header; ~~arrow on the left edge for the row; top-left corner for the whole table.~~
- • Ctrl+A once selects the data, twice includes the headers; Ctrl+Space selects the column, Shift+Space the table row.

Limits worth knowing

- With a table on the sheet, Subtotals (Data ► Subtotal), Custom Views and the legacy shared-workbook mode are unavailable.
- Group/Outline works only on sheet rows, not "inside" the table's logic.

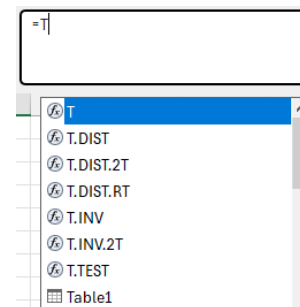
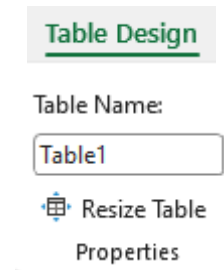
Where the table name appears?

It doesn't appear anywhere on the sheet itself: the name is only visible in the interface, in three main places.

1. **Table Design tab** (in Excel 2010–2019 "Table Tools > Design"): it only appears when the active cell is inside the table. At the far left, in the Properties group, there is the "Table Name" box showing Table1. It is also the only place where we can rename it: in Name Manager the table name is listed (with a different icon from the other names), but the Edit and Delete buttons are disabled.
2. **Name Box** (to the left of the formula bar): opening the drop-down list we find Table1 among the names; selecting it, or typing it and pressing Enter, selects the table's data body. Conversely, if we select exactly the data body (one click on the arrow in the top-left corner of the table), the box shows Table1.
3. **In formulas**: typing `=Tab` makes Formula AutoComplete suggest Table1 with the table icon, and we can use it in structured references, for example `=SUM(Table1[Amount])`.

We then find it wherever the table is used as a source: in the "Table/Range" field when creating a PivotTable, as the query name in Power Query, in charts, in VBA with `ListObjects("Table1")`.

One useful detail: the name Table1 refers to the data body only, without the header row (Name Manager shows this in "Refers to"). To include the headers as well we use `Table1[#All]`.



How to convert an Excel Table back to a range

The data are preserved; only the Table structure is removed.

1

Select the Table

Click any cell inside the Excel Table.

2

Open Table Design

The Table Design tab appears on the Ribbon.

3

Convert to Range

Choose Table Design → Convert to Range.

4

Confirm

When Excel asks for confirmation, click Yes.

What changes?

Excel Table**Normal range****Data remain**

Values and formulas are preserved.

Formatting

Table formatting usually remains.

Table features

Automatic expansion and structured Table behavior are removed.

Range → Ctrl+T → Excel Table → Table Design > Convert to Range → Range

Data types in Data Science, regardless Excel: meaning comes before formatting

The type determines which operations are meaningful — and which are nonsense.

Numeric

Measured quantities

Balance, age, income, interest rate

Categorical

A predefined number of values

Labels / groups

Segment, product type, branch

Boolean / binary

A special case of Categorical, with just 2 values

Two states

Default yes/no, active/inactive

Date / time

Time coordinates

Transaction date, month-end

Text

Free-form strings

Comments, descriptions, notes

Identifier

Keys, not quantities

CustomerID, IBAN, account number

A CustomerID that contains digits (eg, 4500) is still an identifier, not a numeric measurement.

Numeric data: discrete and continuous

Both are numeric, but the measurement scale and distribution may differ.

Discrete numeric variables

Countable values, often integers.

Examples: number of products, missed payments, complaints.

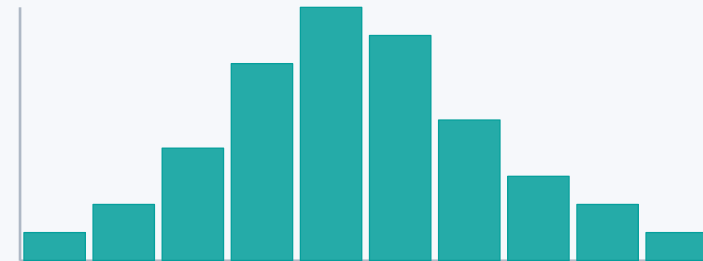


Discrete: Products per customer

Continuous numeric variables (with decimal point)

Measurements on a scale.

Examples: balance, income, amount, duration, interest rate.



Continuous: Account balance

```
=AVERAGE (Table1[Balance])
```

```
=MEDIAN (Table1[Balance])
```

```
=STDEV.S (Table1[Balance])
```

Categorical data: nominal, ordinal and binary

Categories can be labels, ordered levels or two-state flags.

Nominal

No intrinsic order

Region: North / South
Product: Card / Mortgage / Loan

Ordinal

Meaningful order

Risk grade: Low / Medium / High
Satisfaction: 1...5

Binary

Exactly two states

Default: Yes / No
Active: 1 / 0

Excel implications

- Use Data Validation lists to control allowed categories.
- Use PivotTables to compare counts, averages and rates across groups.
- Do not average arbitrary codes (e.g., Region 1, 2, 3).

Data types: qualitative and quantitative

A practical taxonomy for structured (tabular) data — with examples and software mappings.

VARIABLES IN TABULAR DATA

QUALITATIVE DATA

aka categorical

Values are labels or classes. **The set of possible values is often finite and relatively small.** Numeric operations such as sums or divisions are not meaningful.

NOMINAL

Unordered categories

No intrinsic ranking between values.

Examples: ZIP code, eye color, region, product type

ORDINAL

Ordered categories

Values have an order, but gaps are not numeric distances.

Examples: rating, risk class, customer tier

QUANTITATIVE DATA

aka numeric

Values represent quantities. Arithmetic and distances are meaningful. **The theoretical set of possible values may be very large or infinite.**

DISCRETE

Counts

Typically integer-valued observations.

Examples: # children, # transactions, # defects

CONTINUOUS

Measurements

Can take any value within an interval, conceptually.

Examples: temperature, height, balance, interest rate

Software mapping in Excel

[See the next slide](#)

Important: A numeric-looking code is not necessarily numeric data. ZIP code and Customer_ID are identifiers.

Key idea: data type is about meaning and valid operations — not only about how a value is formatted in Excel.

Statistical Data Types vs Excel Data Types

Excel stores values using technical cell types. The analyst assigns the statistical meaning.

Statistical data type	Meaning	Typical Excel format	Example
Nominal categorical	Labels with no inherent order	Text	Region = "Lombardy"
Ordinal categorical	Categories with a meaningful order	Text or coded Number	Low / Medium / High
Discrete numeric	Counts; integer-valued measurements	Number (integer value)	Num_Products = 4
Continuous numeric	Measurements on a continuous scale	Number (decimal value)	Balance = 5,823.47
Binary / Boolean	Two possible states	TRUE/FALSE or 0/1	Defaulted = FALSE
Date / Time	Temporal values	Date/Time*	13/09/2026
Identifier	A code, not a quantity	Usually Text	Customer_ID = "CL00015"
Missing value	Unknown / not recorded	Blank cell	empty cell

* Excel stores dates/times internally as serial numbers, but formats them as dates/times.

KEY IDEA

Excel data type ≠ statistical data type

EXAMPLE 1 — ORDINAL CODES

1 = Low 2 = Medium 3 = High

Excel sees numbers; the analyst sees ordered categories. An average may be mathematically possible but analytically questionable.

EXAMPLE 2 — IDENTIFIERS

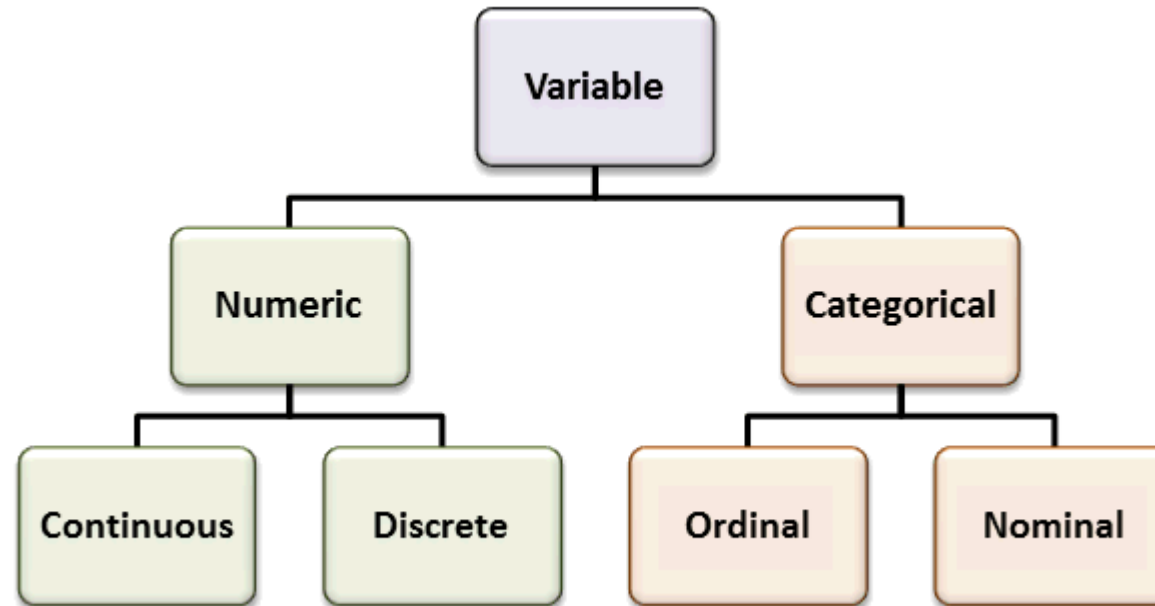
ZIP = 10121 Account = 000173

They look numeric, but they are labels. Store them as Text when leading zeros or code semantics matter.

Analyst rule: determine the variable’s meaning first; only then decide the Excel representation and the valid operations.

Numerical vs Categorical Data Types

Not every column that looks like a number should be treated as a number.



Dates, text and identifiers need special care

Not every column that looks like a number should be treated as a number.

Example	Semantic type	Appropriate analysis
2026-08-31	Date	Sort, group by month/quarter, compute intervals
C100245	Identifier	Join / lookup key — never average it
0012345678	Identifier stored as text	Leading zeros matter
“Mortgage renewal”	Text	Search, classify, count terms
12.5%	Numeric rate	Store as 0.125; display as percentage

Formatting changes appearance; it does not automatically correct the underlying data type.

VLOOKUP and HLOOKUP in Banking

Retrieve related information from a reference table using a lookup key.

In Excel in italiano:

- VLOOKUP → CERCA.VERT
- HLOOKUP → CERCA.ORIZZ

VLOOKUP — Vertical Lookup

Search direction: Searches for the key in the FIRST COLUMN and returns a value from another column.

Banking example: Use Account Code C0004 to retrieve Product Name, Risk Category and Annual Fee.

Formula: =VLOOKUP(A2,\$G\$2:\$J\$6,2,FALSE)

Typical use: Product master data, branch codes, account types, customer categories.

HLOOKUP — Horizontal Lookup

Search direction: Searches for the key in the FIRST ROW and returns a value from another row.

Banking example: Use Rating Code C to retrieve the applicable Interest Rate and Credit Limit.

Formula: =HLOOKUP(B2,\$F\$2:\$J\$4,2,FALSE)

Typical use: Rate grids, rating bands, monthly/quarterly tables arranged horizontally.

Key idea: VLOOKUP = vertical reference table; HLOOKUP = horizontal reference table. FALSE requests an exact match.

- VLOOKUP:**

Bank_Account_Product_Lookup.csv

"The table on the left represents bank transactions. Each row is a transaction and contains only the Account Code and the Transaction Amount.

The table on the right is a reference table that describes each Account Code: it tells us the corresponding Product Name, Risk Category, and Annual Fee.

VLOOKUP uses the Account Code as the key to connect the two tables and automatically add the correct product information to each transaction."

This table design (two separate tables) is required to storage space and avoid harmful redundancy of data (**data table normalization**)

The table resulting from lookup

Account Code	Transaction Amount (EUR)	Product Name	Risk Category	Annual Fee (EUR)		Lookup Table			
C0001	1250	Current Account	Low	60		C0001	Current Account	Low	60
C0004	8200	Mortgage	Medium	250		C0002	Savings Account	Low	0
C0003	35000	Personal Loan	Medium	120		C0003	Personal Loan	Medium	120
C0002	5400	Savings Account	Low	0		C0004	Mortgage	Medium	250
C0005	980	Credit Card	High	45		C0005	Credit Card	High	45

Data Table (Transactions)

Lookup Table

Both the lookup tables are *reference tables* (by definition) – the first is organised by rows, the second by columns

- HLOOKUP:**

Credit_Rating_Terms_Lookup.csv

"Credit ratings are arranged horizontally across the top of the table. HLOOKUP uses the customer's rating code to retrieve the corresponding interest rate, credit limit, or other lending conditions."

The table resulting from lookup

Customer	Rating Code	Interest Rate	Credit Limit (EUR)		Horizontal	Lookup Table			
Client A	A	3.5%	50000		A	B	C	D	E
Client B	C	6.0%	20000		3.5%	4.5%	6.0%	8.0%	10.0%
Client C	B	4.5%	35000		50000	35000	20000	10000	5000
Client D	D	8.0%	10000						
Client E	E	10.0%	5000						

Data Table

Lookup Table Table

aka, master data

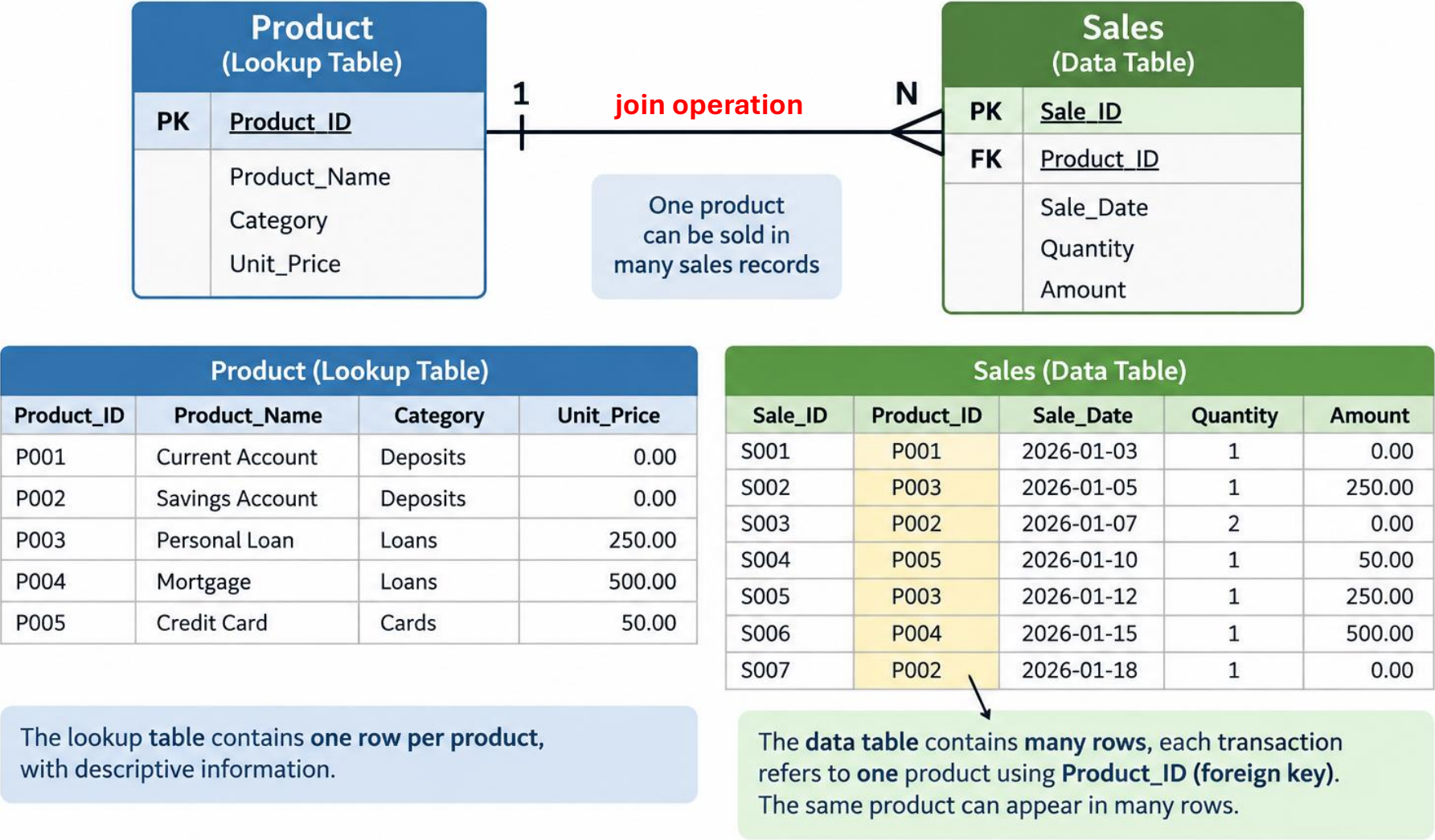
"VLOOKUP is useful when reference data is organized vertically by rows, while HLOOKUP is useful when categories are arranged horizontally across columns."

VLOOKUP in Banking — Enriching Transactions with Product Information							
Business scenario: the transaction table contains only an Account Code and a Transaction Amount. The Product Master table stores the business attributes associated with each Account Code. VLOOKUP uses the Account Code as a key to retrieve Product Name, Risk Category, and Annual Fee.							
Account Code	Transaction Amount (EUR)			Account Code	Product Name	Risk Category	Annual Fee (EUR)
C0001	1,250.00			C0001	Current Account	Low	60.00
C0004	8,200.00			C0002	Savings Account	Low	0.00
C0003	35,000.00			C0003	Personal Loan	Medium	120.00
C0002	5,400.00			C0004	Mortgage	Medium	250.00
C0005	980.00			C0005	Credit Card	High	45.00
Account Code	Transaction Amount (EUR)	Product Name	Risk Category	Annual Fee (EUR)			
C0001	1,250.00	Current Account	Low	60.00			
C0004	8,200.00	Mortgage	Medium	250.00			
C0003	35,000.00	Personal Loan	Medium	120.00			
C0002	5,400.00	Savings Account	Low	0.00			
C0005	980.00	Credit Card	High	45.00			
Formula pattern: =VLOOKUP(lookup value, table array, column index, FALSE) Example: =VLOOKUP(A15,\$E\$7:\$H\$11,2,FALSE)							

HLOOKUP in Banking — Retrieving Lending Conditions from a Rating Matrix									
Business scenario: the customer table contains a Credit Rating for each customer. The Lending Conditions matrix stores ratings horizontally across the first row, with the corresponding interest rate and maximum credit limit underneath. HLOOKUP uses the rating as a key to retrieve those conditions.									
Customer ID	Customer Name	Credit Rating		Metric	AAA	AA	A	BBB	BB
CL001	Northwind Ltd	A		Interest Rate	3.2%	3.6%	4.2%	5.1%	6.7%
CL002	Blue River SA	BBB		Maximum Credit Limit (EUR)	500,000	350,000	250,000	150,000	75,000
CL003	Greenfield SRL	AA							
CL004	Metro Retail Ltd	BB							
CL005	Sunrise Group	AAA							
Customer ID	Customer Name	Credit Rating	Interest Rate	Maximum Credit Limit (EUR)					
CL001	Northwind Ltd	A	4.2%	250,000					
CL002	Blue River SA	BBB	5.1%	150,000					
CL003	Greenfield SRL	AA	3.6%	350,000					
CL004	Metro Retail Ltd	BB	6.7%	75,000					
CL005	Sunrise Group	AAA	3.2%	500,000					
Formula pattern: =HLOOKUP(lookup_value, table_array, row_index, FALSE) Example: =HLOOKUP(C15,\$F\$6:\$J\$8,2,FALSE)									

1 : N Relationship Between a Lookup Table and a Data Table

One record in the lookup table can be related to many records in the data table



VLOOKUP and HLOOKUP **join** two tables in a **1:N relationship** in one direction only, from the N side to the 1 side: each detail row is enriched with the attributes of its master, like a left join, provided the master key is unique. A special case is the **master-detail relationship**, where a detail can only exist if its master exists: Excel does not enforce this referential integrity, but a #N/A returned by the lookup pinpoints exactly the orphan details.

Common data type problems in Excel

Many “analysis errors” are actually type errors.

x Type issues hiding in plain sight				
fx =ISTEXT (C4)				
CustomerID	Date	Income	Rate	Balance
00125	31/08/26	65,000	4.25%	12,500
00126	08/31/26	54.000	0,0425	8,200
00127	text	92,000	4.10%	—
00128	31/08/26	71k	4.35%	165,000

Symptoms

Sort order looks wrong · SUM ignores values · dates do not group.

Checks

ISTEXT · ISNUMBER · DATEVALUE · VALUE · NUMBERTVALUE.

Prevention

Define schema, use Tables, validation, Power Query types and controlled imports.

=ISNUMBER (C2) =ISTEXT (C2) =DATEVALUE (B2)
=NUMBERTVALUE (C2)

Chapter title

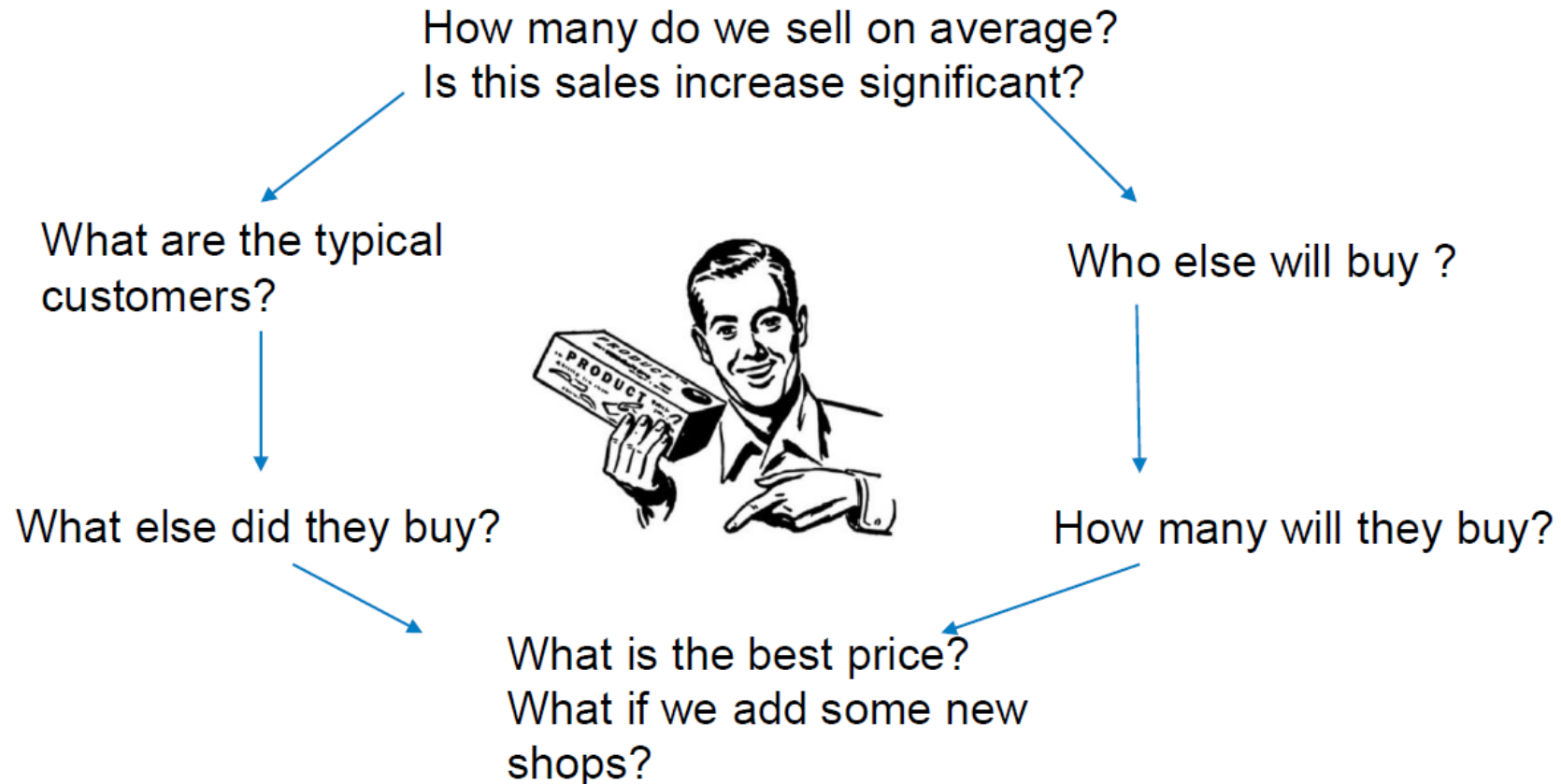


03 – Exploratory Data Analysis (EDA)

Data Science: we always start from a “business problem”

EDA is a disciplined conversation with the dataset: structure → distributions → relationships → problems → hypotheses.

Example: How to sell more of your favorite product?



La tecnica ML di soluzione

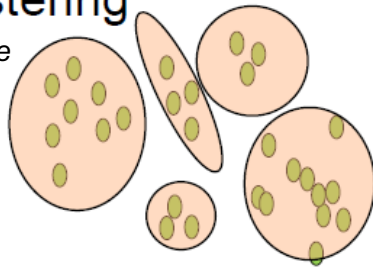
(1:1 posizionale con la slide precedente)

$$Y = f(\text{reddito}, \text{zona}) \\ = k + w_1 \text{reddito} + w_2 \text{zona} + \text{reddito}^2$$

Example: How to sell more of your favorite product?

Clustering

(segmentazione clienti)



Association

Nel marketing:
Market Basket
Analysis, cross-
selling



Reporting & Statistics

(nuovi termini: ML
descrittivo, EDA)

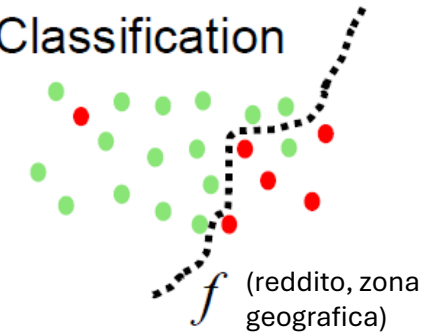


Operations Research

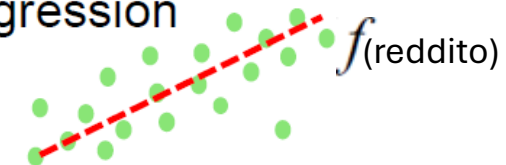


Analisi what-if

Classification



Regression



Exploratory Data Analysis (EDA): learn the data before modelling it

EDA is a disciplined conversation with the dataset: structure → distributions → relationships → problems → hypotheses.

STRUCTURE

rows · columns · types · grain

DISTRIBUTIONS

centre · spread · shape

histograms

RELATIONSHIPS

correlation · groups · trends

DATA ISSUES

missing · outliers ·
inconsistencies

PATTERNS

segments · seasonality ·
anomalies

EDA output is not just charts

It is a set of findings: what the data represents, what looks suspicious, what relationships exist, and what should be investigated next.

EDA = ask → inspect → visualise → quantify → question → repeat

EDA in practice: one graphical example for each of the five lenses

Each lens asks a different question: how the table is organised, how values are distributed, how variables move together, where the problems are, and what broader regularities emerge.

STRUCTURE

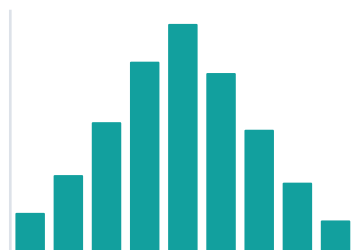
A table with fields and records

ID	Age	Region	Balance
C0	32	Nor	5,2
C0	41	We	8,1
C0	29	So	3,4
C0	55	Ea	9,7
04		st	80

Example question:
Do we have one row per customer, clear column names and meaningful data types?

DISTRIBUTION

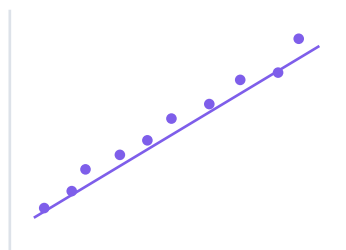
Histogram of a numeric variable



Example question:
How are transaction amounts distributed? Is the shape symmetric, skewed or multi-modal?

RELATIONSHIPS

Scatter plot between two variables



Example question:
Do income and account balance move together? A scatter plot reveals direction and strength.

DATA ISSUES

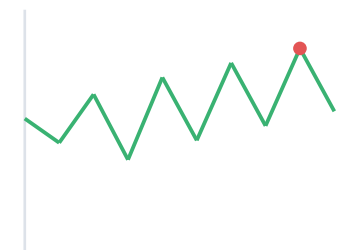
Missing values and outliers

Age	Income	Score
34	28,00	71
41	MV	69
39	31,50	70
36	999,999	72

Example question:
Where are the blanks, errors or extreme values that could distort averages, correlations, charts, and even predictive models?

PATTERNS

Trend / seasonality / anomaly

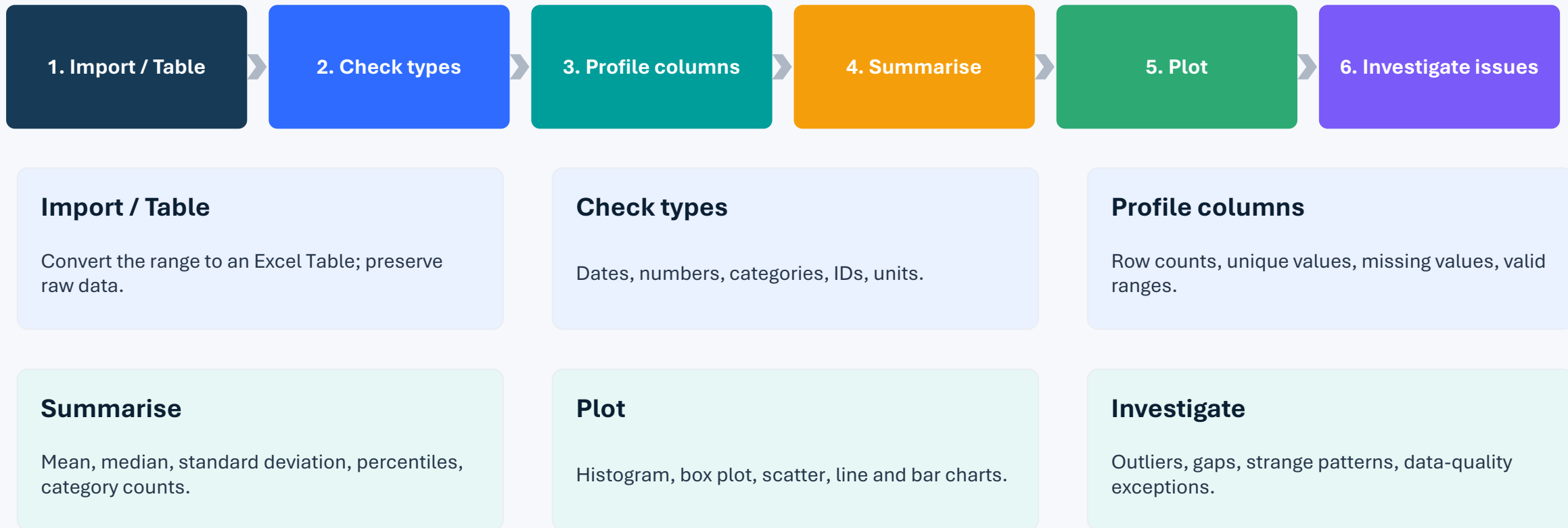


Example question:
Is there a repeated monthly pattern, an upward trend or a point that breaks the usual behaviour?

EDA = structure + distributions + relationships + data issues + patterns

A practical EDA workflow in Excel?

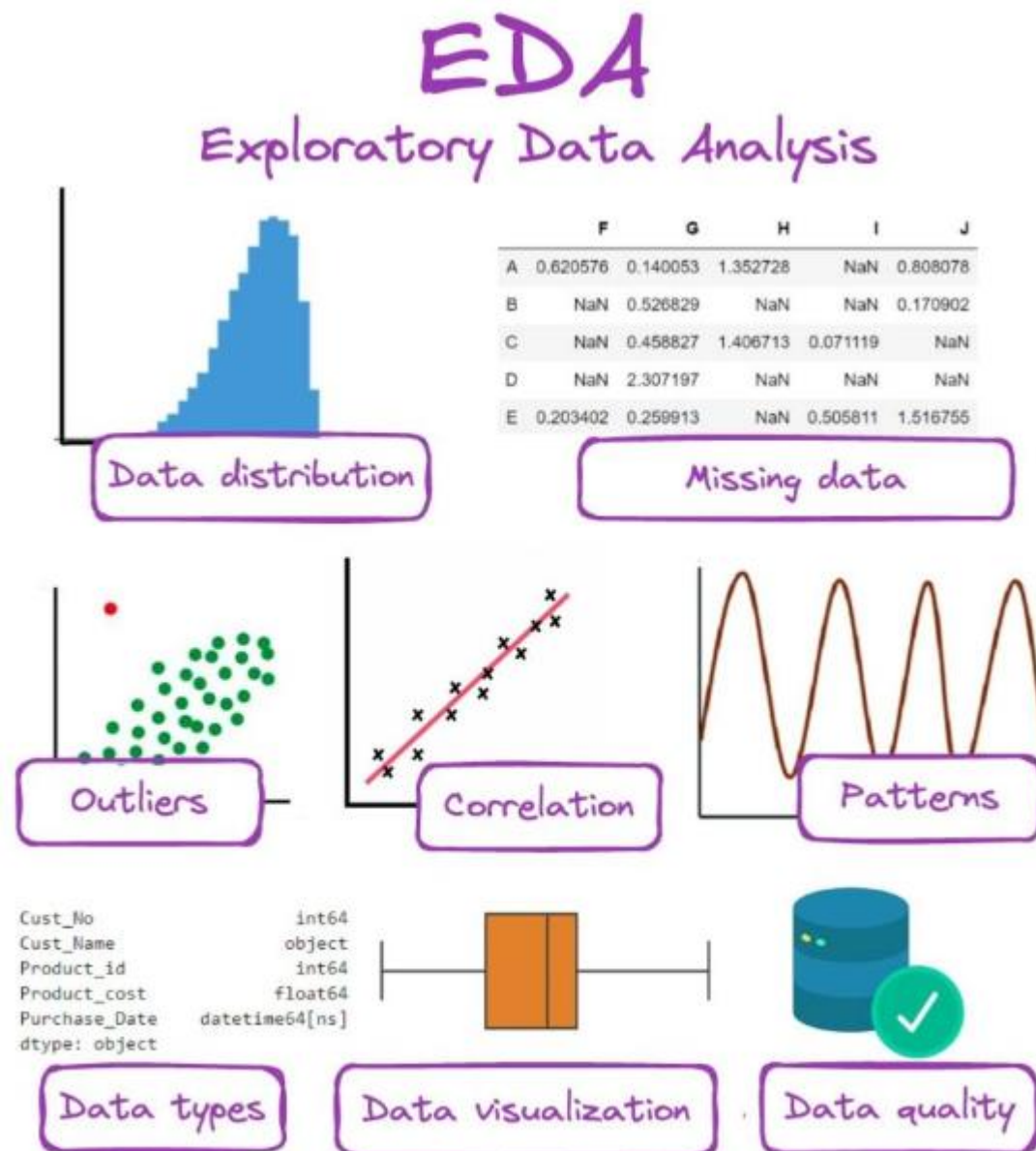
Use a repeatable sequence so important checks are not skipped.



Best practice: keep “raw”, “clean”, “analysis” and “output” areas separate.

A better EDA workflow in Excel!

Use a repeatable sequence so important checks are not skipped.



CSV files: simple tabular data in plain text

CSV stands for Comma-Separated Values: a lightweight format for storing and exchanging structured data.

customers.csv

```
Customer_ID, Age, Income, Balance, Default  
CL0001, 42, 65000, 12500, No  
CL0002, 35, 54000, 8200, No  
CL0003, 61, 92000, 45500, No  
CL0004, 29, 45000, 3200, Yes
```

Each line is a row. Separators divide fields into columns. The first row usually contains column names.

Plain text

very convenient for import / export

No formulas, charts, formatting or multiple worksheets — only values.

Tabular structure

Rows represent observations; fields represent variables.

Delimiter matters

Despite the name, CSV files may use commas, semicolons or other delimiters depending on locale and source.

CSV is a data format, not a spreadsheet workbook.

Why CSV is useful in Data Science

CSV is widely used as a simple interchange format between analytical tools, databases and programming environments.

Portable

Readable by Excel, Python, R, SQL tools, BI platforms and many other applications.

Transparent

The content can be inspected with a text editor: there is little hidden structure.

Easy to exchange

Useful for exporting datasets from systems and moving tabular data between tools.

Automation-friendly

Straightforward to read and write in scripts and data pipelines.

Lightweight

No workbook features are stored, so the format focuses on the data itself.

Good for datasets

A natural representation for rows × columns used in EDA and Machine Learning.

Typical Data Science flow: source system → CSV → Excel / Python / R → analysis

CSV

CSV ↔ Excel: how to convert them in both directions

Import a CSV when you want to analyse it in Excel; export to CSV when you need a simple, portable data file.

CSV → Excel

1. In Excel, choose Data → From Text/CSV.
2. Select the .csv file.
3. Check delimiter, encoding and the previewed columns.
4. Choose Load to bring the data into Excel.
5. Save as .xlsx if you want to preserve Excel features.

Excel → CSV

1. Keep the worksheet you want to export active.
2. Choose File → Save As.
3. Select CSV UTF-8 (Comma delimited) (*.csv) when appropriate.
4. Save the file and confirm Excel's warnings.
5. Re-open or inspect the CSV if delimiter/encoding is critical.

bank_customers_missing.csv
CSV UTF-8 (Comma delimited) (*.csv)
Excel Workbook (*.xlsx)
Excel Macro-enabled Workbook (*.xlsm)
Excel Binary Workbook (*.xlsb)
Excel 97-2003 Workbook (*.xls)
CSV UTF-8 (Comma delimited) (*.csv)
XML Data (*.xml)
Single File Web Page (*.mht;*.mhtml)
Web Page (*.htm;*.html)
Excel Template (*.xltx)
Excel Macro-enabled Template (*.xltm)
Excel 97-2003 Template (*.xlt)
Text (Tab delimited) (*.txt)
Unicode Text (*.txt)
XML Spreadsheet 2003 (*.xml)
Microsoft Excel 5.0/95 Workbook (*.xls)
CSV (Comma delimited) (*.csv)
Formatted Text (Space delimited) (*.prn)
Text (Macintosh) (*.txt)
Text (MS-DOS) (*.txt)
CSV (Macintosh) (*.csv)
CSV (MS-DOS) (*.csv)
DIF (Data Interchange Format) (*.dif)
SVLK (Symbolic Link) (*.slk)
Excel Add-in (*.xlam)
Excel 97-2003 Add-in (*.xla)
PDF (*.pdf)
XPS Document (*.xps)
Strict Open XML Spreadsheet (*.xlsx)
OpenDocument Spreadsheet (*.ods)

Important: CSV stores one flat table only — formulas, formatting, charts, PivotTables and additional worksheets are not preserved.

CSV files across locales: when commas stop being simple

A CSV file is plain text. The delimiter, decimal symbol, dates and encoding may be interpreted differently by Excel depending on regional settings.

KEY IDEA Never assume that “CSV” means the same parsing rules on every computer. Import settings must be explicit.

1 What can change with locale?

- Field delimiter** Comma (,), semicolon (;), tab, etc.
- Decimal symbol** 3.75 in EN/US vs 3,75 in many EU locales
- Thousands separator** 25,900.50 vs 25.900,50
- Date interpretation** 03/04/2026 can mean Mar 4 or Apr 3
- Text encoding** UTF-8 avoids many accented-character problems

2 Example: the same data, different conventions

Source CSV created in an English/US environment

```
Customer_ID,Income,Rate,Region
C001,25900.50,3.75,Turin
C002,17200.00,4.10,Milan
```



Double-click / automatic opening

POSSIBLE RESULT

- all fields in one column
- decimal values split incorrectly
- numbers stored as text
- dates silently reinterpreted

Explicit import settings

SAFE RESULT

- delimiter set explicitly
- decimal locale set correctly
- columns typed as expected
- row/column counts validated

3 Safe Excel import workflow

1

Data → From Text/CSV

Do not rely on double-click when locale is uncertain.

2

Set delimiter explicitly

Choose comma, semicolon, tab, etc.

3

Check File Origin / UTF-8

Preserve accented characters and symbols.

4

Set locale for numbers & dates

Especially decimal comma/dot and date order.

5

Validate before analysis

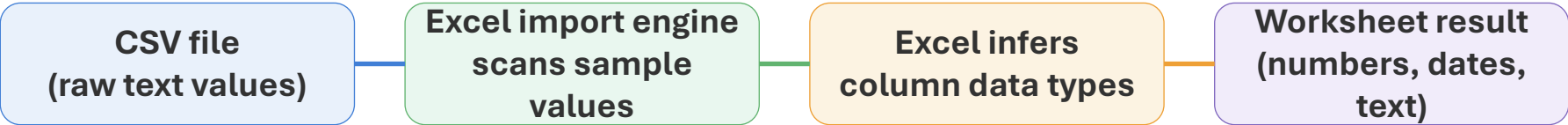
Check column count, data types, totals and sample rows.

For shared workflows: document the delimiter, encoding and locale.

CSV import in Excel: data type inference

Why the same CSV file can be interpreted differently across tools, locales and workflows

WHAT EXCEL DOES



Typical inference examples

CSV values	Inferred as	Possible issue
"00123"	Number	Leading zeros lost
"13/09/2026"	Date	Locale ambiguity
"1,234"	Number/Text	Decimal vs thousands
"TRUE"	Logical/Text	Depends on parser
"NA" / blank	Missing/Text	May not mean the same thing
"CL00015"	Text	Usually OK

Key idea: a CSV file stores text separated by delimiters; data types are reconstructed during import.

Why this matters

The same column may be imported as text, number or date. Wrong inference can distort sorting, filtering, formulas and charts. IDs, ZIP codes and account numbers can be damaged by numeric conversion. Cross-language imports may change decimal separators and date interpretation.

Safer Excel workflow
Use Data → From Text/CSV instead of double-clicking the file.
Review delimiter, file origin / encoding, locale, and detected data type.
Force Text for identifiers; verify dates, decimals and missing-value markers.

Takeaway: CSV files do not carry rich type metadata. Excel infers types during import, so analysts must review the import settings before trusting the data.

First look: create a dataset profile

Before any deep analysis, answer a small set of basic questions.

x

Column profile

fx

=COUNTBLANK (Table1[Income])

Field	Type	Missing	Unique	Min	Max
Age	Numeric	0	41	18	86
Income	Numeric	37	1,204	12k	420k
Segment	Category	0	4	—	—
Balance	Numeric	4	8,951	-1.2k	650k
Default	Binary	0	2	0	1

How many rows?

ROWS, COUNTA or Table totals.

How many unique values?

UNIQUE + ROWS (modern Excel) or PivotTable.

What is missing?

COUNTBLANK, ISBLANK, filters and conditional formatting.

=ROWS (Table1)

=COUNTA (Table1[CustomerID])

=COUNTBLANK (Table1[Income])

Descriptive statistics: centre, spread and shape

Summary statistics compress a distribution into a few interpretable numbers.

Mean

Average level
=AVERAGE (range)

Median

Middle value
=MEDIAN (range)

Std. deviation

Typical spread
=STDEV.S (range)

Min / Max

Observed range
=MIN (range) / =MAX (range)

Quartiles

25th / 50th / 75th percentiles
=QUARTILE.INC (range , k)

Important

Mean and standard deviation can be heavily affected by outliers. Always compare with median and quartiles.

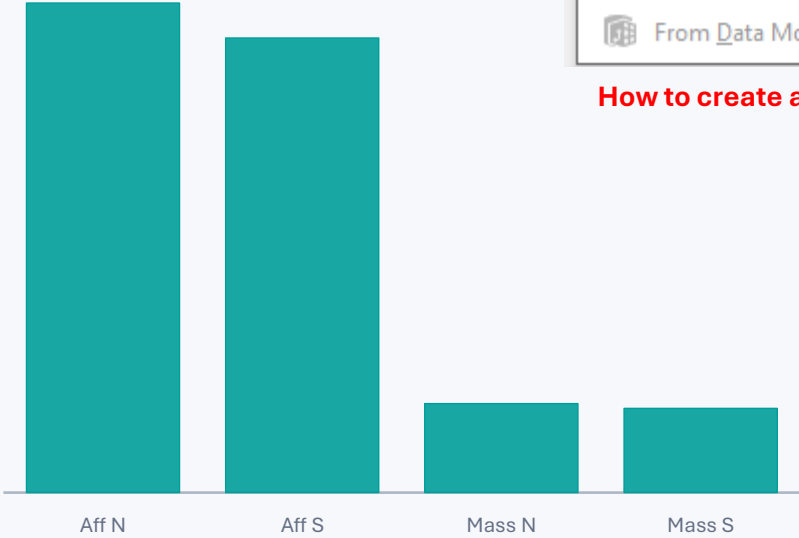
Age		
Mean	46.03	
Standard Error	0.46	
Median	47.00	
Mode	52.00	
Standard Deviation (σ)	14.51	
Sample Variance	210.53	
Kurtosis	0.28	
Skewness	-0.20	
Range	85.00	
Minimum	0.00	
Maximum	85.00	
Sum	44928.00	
Count	976.00	
Confidence Level(95.0%)	0.91	
CI for population mean is		
	45.12	46.94

The “Descriptive Statistics” option in Excel Analysis ToolPak can generate a compact summary table.

PivotTables: fast multivariate EDA without writing formulas

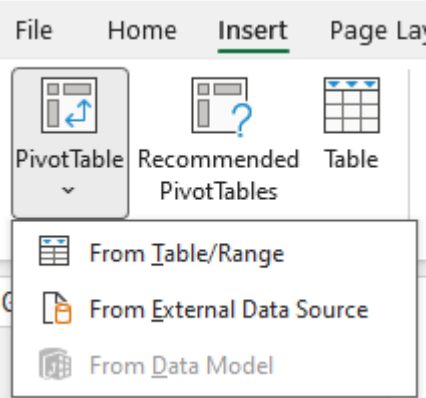
PivotTables help answer “how does a metric differ across groups?”

x PivotTable output				
fx Average of Balance / Count of CustomerID				
Segment	Region	Customers	Avg Balance	Default %
Affluent	North	1,240	46,200	0.8%
Affluent	South	980	42,900	1.1%
Mass	North	4,650	8,400	2.9%
Mass	South	4,210	7,950	3.4%



Average balance by segment & region

PivotTables are a bridge between raw tables and clear business questions.



How to create a pivot table

Sorting, filtering and conditional formatting are EDA tools

Simple spreadsheet operations often reveal data issues faster than complex methods.

xFilter & inspect

fx=

=SORT (Table1[Balance], , -1)

Customer	Income	Balance	DaysPastDue
C1001	65,000	12,500	0
C1002	54,000	8,200	0
C1003	92,000	45,500	12
C1004		3,200	0
C1005	71,000	165,000	97
C1006	58,000	9,700	0

Sort

Find top/bottom values and suspicious extremes.

Filter

Inspect one segment, branch, category or exception at a time.

Conditional formatting

Highlight blanks, duplicates, thresholds and unusual values.

Freeze panes / Table headers

Keep context while scanning many rows.

Tip: “show me the 20 largest values” is often a very good EDA question.

Conditional Formatting in Excel: make important patterns visible

Conditional Formatting changes how cells look when rules are met — helping analysts spot exceptions, thresholds and data-quality issues quickly.

WHAT IT IS

A visual rule applied to cells based on their values or on a formula.

Examples:

- values above a threshold
- blank or duplicate cells
- Data Bars and Color Scales
- Icon Sets for status/risk

It changes appearance, not the underlying value.

EXAMPLE

Flag unusual account balances

Customer_ID	Balance	Status
C001	1,200	OK
C002	850	OK
C003	15,900	HIGH
C004		MISSING
C005	670	OK
C006	22,300	HIGH

Exceptions become visible without changing the dataset.

HOW TO USE IT

Home → Conditional Formatting

Main rule types:

- Highlight Cells Rules / Top-Bottom
- Data Bars / Color Scales / Icon Sets
- New Rule / Use a formula

Formula-based rules

=ISBLANK(\$M2) → missing value

=\$M2>10000 → threshold

Common Data Science uses

Missing values

Outliers

Thresholds

Duplicates

Trends / magnitude

Takeaway: Conditional Formatting is a visual diagnostic layer — useful for seeing patterns quickly, but it does not clean or modify the data.

The dialog classifies rules **by criterion** (how the test is defined), the menu **by effect and by use** (what we want to obtain and how quickly).

Conditional Formatting in Excel: the rule types



Type in the dialog	Corresponding menu item	Extra options versus the menu
Format all cells based on their values	Data Bars, Colour Scales, Icon Sets	thresholds as number/percentile/formula, 2-colour scale, bar only
Only format cells that contain	Highlight Cells Rules (Greater Than, Between, Equal To, Text that Contains, A Date Occurring)	Not Between, blanks/no blanks, errors/no errors
Only format top or bottom ranked values	Top/Bottom Rules (Top/Bottom 10 Items, 10 %)	any N
Only format values that are above or below average	Top/Bottom Rules (Above/Below Average)	1, 2 or 3 standard deviations above/below the mean, "equal or above"
Only format unique or duplicate values	Highlight Cells Rules ▶ Duplicate Values	—
Use a formula to determine which cells to format	none: only through New Rule...	anything that can be written as a formula

Conditional Formatting - Examples

Two files ready.

Workbook `Credit_ISLR_conditional_formatting.xlsx` — two sheets:

- **Customers:** the 400 customers in the table `tblCustomers` (A1:L401) with 12 rules, one for each category of the Conditional Formatting menu:
 - Highlight Cells: *Greater Than* 10,000 on Limit, *Duplicate Values* on Limit (border only, so it coexists with the fill), *Between* 18–30 on Age, *Equal To* "Yes" on Student
 - Top/Bottom: *Above Average* on Education, *Top 10 Items* on Balance
 - *Data Bars* on Income, *Color Scale* on Rating, *Icon Set* (traffic lights with numeric thresholds, reversed order) on Cards
 - Formulas (New Rule ▶ Use a formula):
`=B2>AVERAGE($B$2:$B$401)+2*STDEV($B$2:$B$401)` on Income, `=L2=0` and `=AND($I2="Yes", $L2>1000)` on the whole row
- **Rules:** for each rule the menu path, the dialog settings, the equivalent formula, the format applied and the count of matching cells computed with live formulas (COUNTIF, LARGE, SUMPRODUCT...).

Guide `Conditional_Formatting_Guide_Excel2024.md` (in English, "we" form): importing the CSV with Power Query or Ctrl+T, how conditional formatting works (selection, `$` references, priority), the 12 operational steps, the Rules Manager, a summary table and common pitfalls.

Three notes:

- Two clean-ups on the CSV: the first unnamed column (a duplicate of ID) was dropped and the leading space in `" Male"` was removed.
- The data in the file is embedded, not linked through a query: the Power Query connection cannot be recreated from outside Excel, but the steps to build it are in the guide (option A).
- The rule formulas use A1 references because conditional formatting does not accept structured references; the z-score rule is saved with `STDEV` (equivalent to `STDEV.S`) for compatibility.

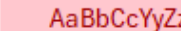
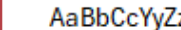
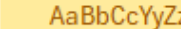
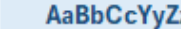
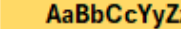
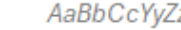
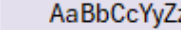
Conditional Formatting - Examples

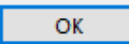
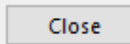
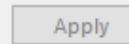
 Manage Rules...

Conditional Formatting Rules Manager

Show formatting rules for: This Table

 New Rule...
  Edit Rule...
  Delete Rule
  Duplicate Rule
 


Rule (applied in order shown)	Format	Applies to	Stop If True
Data Bar		=B\$2:B\$401 	☐
Cell Value > 10000		=C\$2:C\$401 	☐
Duplicate Values		=C\$2:C\$401 	☐
Graded Colour Scale		=D\$2:D\$401 	☐
Icon Set		=E\$2:E\$401 	☐
Cell Value between 18 and 30		=F\$2:F\$401 	☐
Above Average		=G\$2:G\$401 	☐
Cell Value = "Yes"		=I\$2:I\$401 	☐
Top 10		=L\$2:L\$401 	☐
Formula: =B2 > AVERAGE(B\$2:B\$401) + 2*STDEV(B\$2:B\$401)		=B\$2:B\$401 	☐
Formula: =\$L2=0		=A\$2:L\$401 	☐
Formula: =AND(\$I2="Yes";\$L2>1000)		=A\$2:L\$401 	☐

 OK
  Close
  Apply

- Format all cells based on their values
- Only format cells that contain
- Only format unique or duplicate values
- Format all cells based on their values
- Format all cells based on their values
- Only format cells that contain
- Only format values that are above or below average
- Only format cells that contain
- Only format top or bottom ranked values
- Use a formula to determine which cells to format
- Use a formula to determine which cells to format
- Use a formula to determine which cells to format

Conditional Formatting - Examples

Built-in rule — New Rule... ▶ *Only format cells that contain* ▶ in the first drop-down, instead of *Cell Value*, we choose **Blanks** (or *No Blanks*, and next to them *Errors / No Errors* for #N/A and the like). It has no item in the galleries, so it is only visible from New Rule. Internally Excel stores it as `=LEN(TRIM(B2))=0`, so it also catches cells that contain only spaces.

Formula rule, more flexible, applied to the table body A2:L401:

- `=ISBLANK(A2)` → truly empty cells only; a cell holding an empty string `""` (typical after an import or a formula) is *not* caught;
- `=LEN(TRIM(A2))=0` → empty cells, `""` and spaces only;
- `=OR(A2="",A2="NA",A2="N/A",A2="-",A2="?")` → "disguised" missing values stored as text placeholders;
- `=AND(A2<>"",NOT(ISNUMBER(B2)))` on numeric columns → text where a number is expected (e.g. "n.a." imported into Income);
- `=COUNTBLANK($A2:$L2)>0` → highlights the **entire row** when at least one value is missing (COUNTBLANK counts both empty cells and `""`).

The Credit dataset has no missing values, so on the current file these rules would colour nothing; for a demo we delete a few cells and the formatting appears immediately. If you like, I can add to the workbook a rule 13 (orange fill on empty cells) and a rule 14 (row with at least one missing value) and update the Rules sheet and the guide.

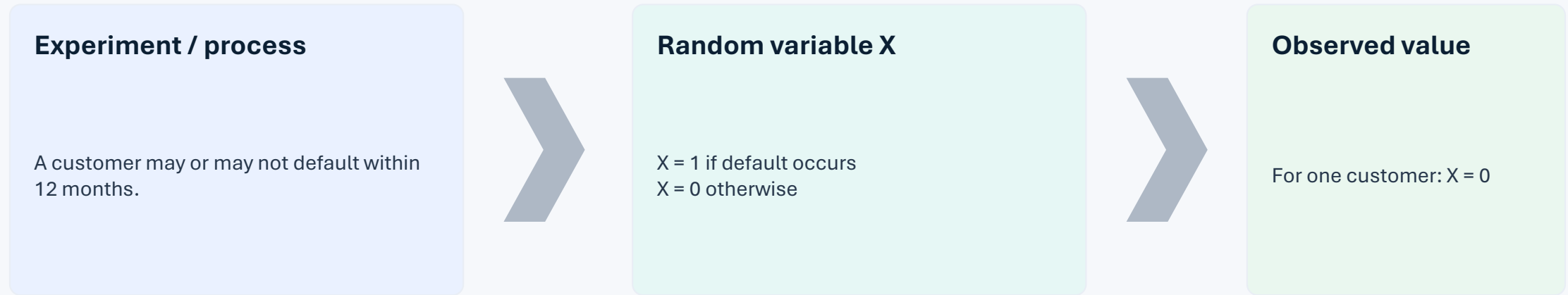
Chapter title



04 – Probability & Statistics

Random variables: a bridge from data to probability

A random variable represents a quantity whose observed value depends on an uncertain outcome.



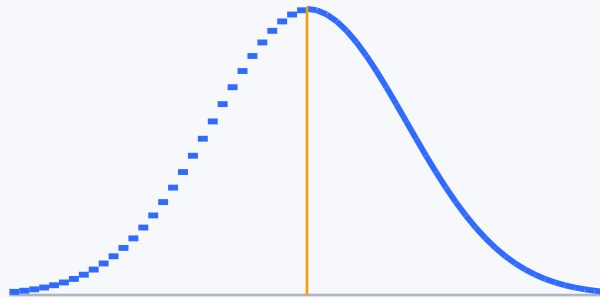
Other examples

Transaction amount · number of late payments · account balance next month · time until loan closure · daily loss.

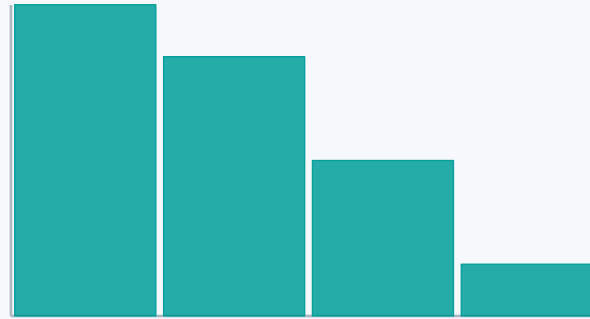
Random does not mean “without structure”. It means the exact outcome is uncertain before observation.

Probability distributions: the “shape” of uncertainty

A distribution describes which values a random variable can take and how likely they are.



Continuous distribution



Discrete distribution

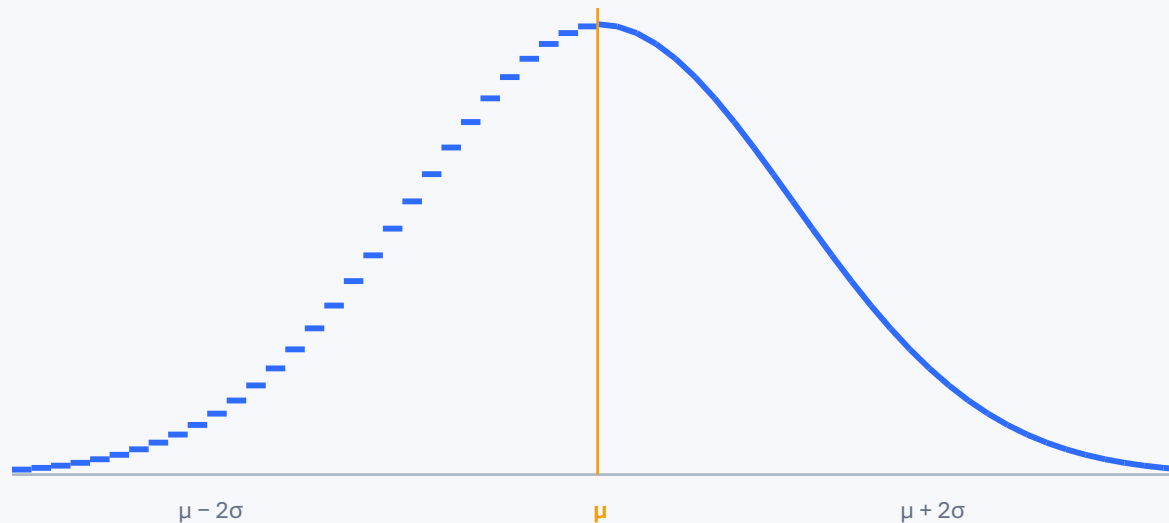


Binary distribution

- Distributions help us reason about typical values, variability and tail risk.
- The empirical distribution comes from observed data; a theoretical distribution is a model.
- EDA asks whether the observed shape is symmetric, skewed, multi-modal, heavy-tailed or irregular.

The Normal distribution: useful model, not universal law

Many measurement errors and aggregate quantities are approximately bell-shaped — but financial data often has skew and heavy tails.



Parameters

Mean μ controls location; standard deviation σ controls spread.

EDA question

Does the histogram look roughly symmetric and bell-shaped?

Banking caution

Losses, balances and transaction amounts are often skewed and may have heavy tails.

Excel examples: `=NORM.DIST(x, mean, stdev, FALSE)` | `=NORM.INV(p, mean, stdev)`

Bernoulli and Binomial: natural distributions for “event / no event”

Binary outcomes appear everywhere in banking: default, fraud flag, churn, approval, response.

Bernoulli random variable

One trial with two outcomes.

$X = 1$ with probability p

$X = 0$ with probability $1 - p$



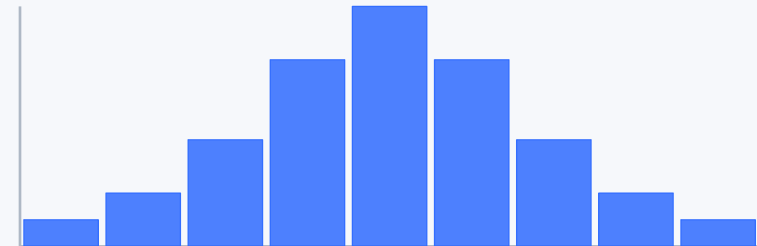
Binomial random variable

Counts how many “successes” occur in n comparable Bernoulli trials.

Example: number of defaults among 100 similar loans.



Bernoulli: one customer



Binomial: count across many customers

Probability vs statistics: opposite directions of reasoning

Probability starts from a model; statistics starts from data.

Probability

MODEL / assumptions



Possible data / outcomes

Statistics

OBSERVED DATA



Model / parameters / inference

EDA sits mainly on the statistics side: use observed data to learn what is plausible, typical and suspicious.

Sampling and uncertainty: a sample is not the population

Any estimate from a sample varies from sample to sample. Statistical thinking quantifies that uncertainty.

Population



Sample



Estimate

Sample mean
Sample default rate
Sample correlation
... each has uncertainty.

- Larger, representative samples usually reduce uncertainty.
- Biased samples can be precise but wrong.
- Always ask: “how was this data selected?”

Chapter title

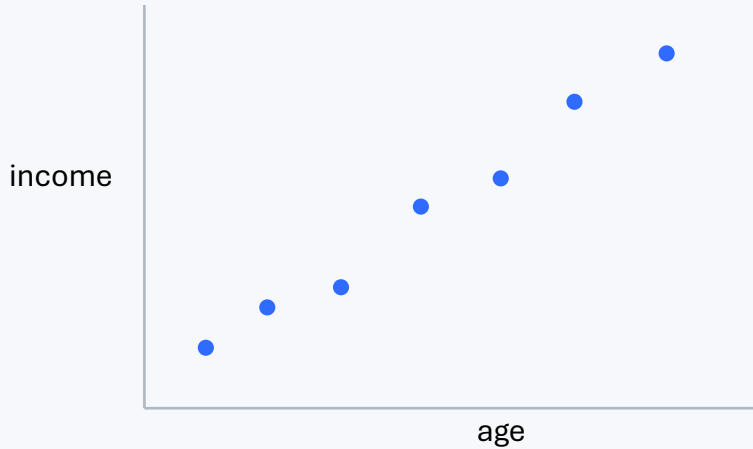


05 - Relationships

Correlation: how two numeric variables move together

Correlation quantifies the strength and direction of a linear relationship.

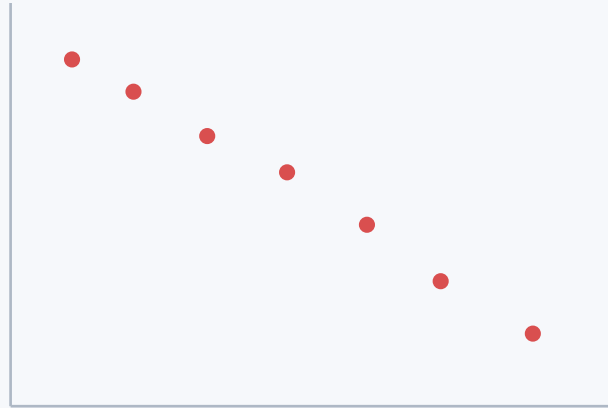
Positive correlation



$$r > 0$$

High values of X tend to occur with high values of Y.

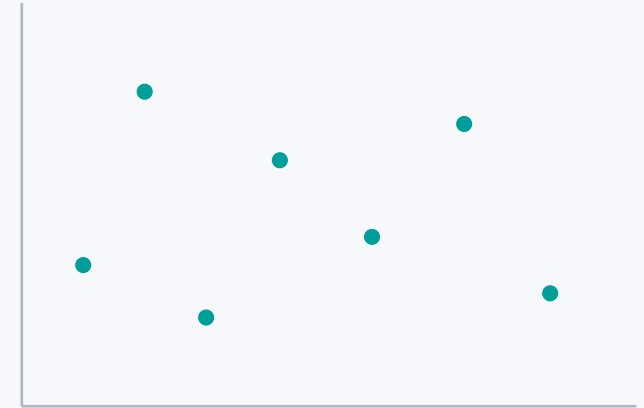
Negative correlation



$$r < 0$$

High values of X tend to occur with low values of Y.

Little linear correlation



$$r \approx 0$$

No strong linear relationship — but nonlinear structure may still exist.

The Pearson correlation coefficient

Pearson r is bounded between -1 and $+1$ and measures linear association.

CORREL computes the **Pearson product-moment correlation coefficient** — the standard linear correlation, identical to Excel's **PEARSON** function (the two exist for historical reasons and return the same value):

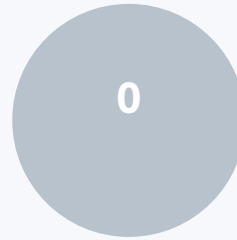
Excel has no built-in Spearman or Kendall. For **Spearman's rank correlation** we rank both columns first and then correlate the ranks:



Perfect negative



Moderate negative



No linear relationship



Moderate positive



Perfect positive

Scale-free

Changing EUR to thousands of EUR does not change r .

Linear

r can miss curves, thresholds and other nonlinear patterns.

Sensitive to outliers

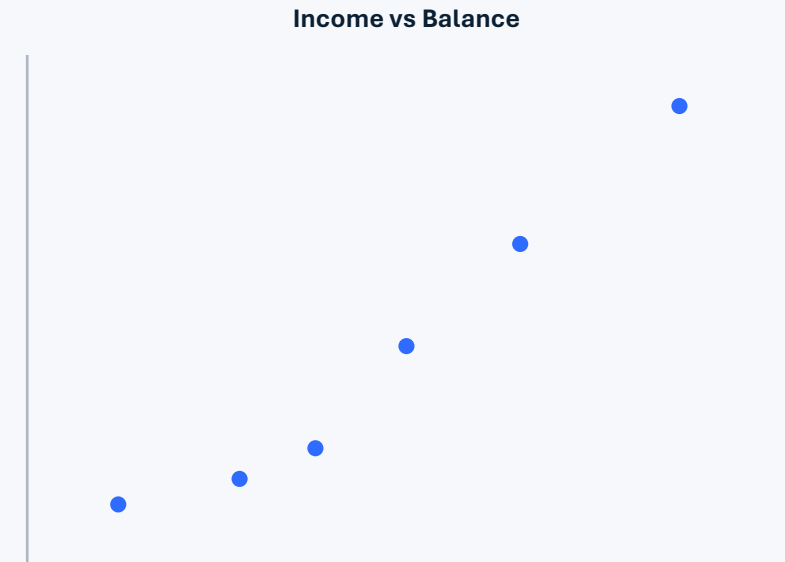
One extreme point can dramatically change the result.

Interpret the coefficient together with a scatter plot.

Correlation in Excel: CORREL + scatter plot

A number is useful; a number plus a plot is much safer.

x Income vs Balance		
fx =CORREL(Table1[Income],Table1[Balance])		
Customer	Income	Balance
C1001	65,000	12,500
C1002	54,000	8,200
C1003	92,000	45,500
C1004	45,000	3,200
C1005	71,000	40,000
C1006	58,000	9,700



=CORREL(Table1[Income], Table1[Balance])

Always check units, missing rows and outliers before interpreting r.

Correlation matrix: many pairwise relationships at once

A correlation matrix is a compact map of linear associations among numeric variables.

	Age	Income	Balance	Products	Arrears
Age	1.00	0.28	0.21	0.15	0.05
Income	0.28	1.00	0.63	0.36	-0.18
Balance	0.21	0.63	1.00	0.42	-0.31
Products	0.15	0.36	0.42	1.00	-0.12
Arrears	0.05	-0.18	-0.31	-0.12	1.00

What to look for

Large absolute values $|r|$ reveal strong linear association.

Diagonal

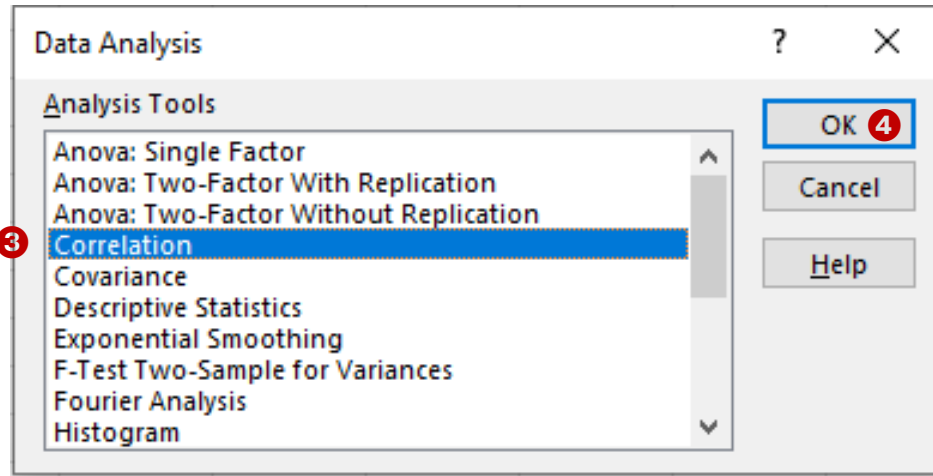
A variable is perfectly correlated with itself: $r = 1$.

Symmetry

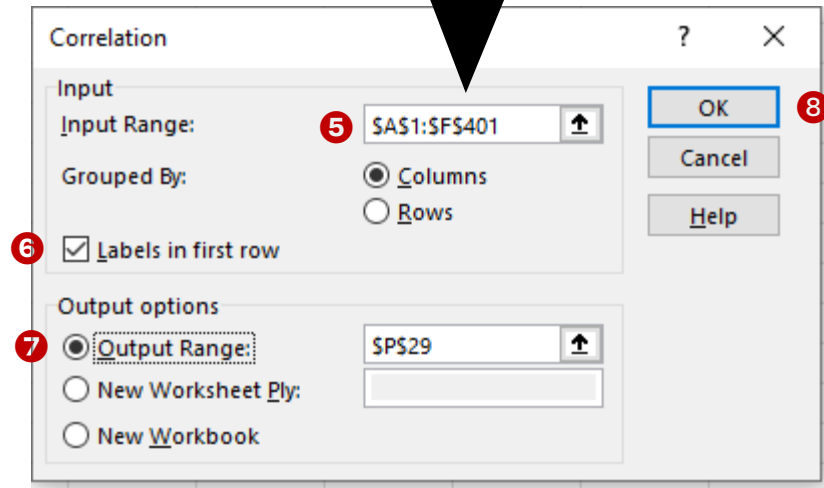
$\text{corr}(X,Y) = \text{corr}(Y,X)$, so the matrix mirrors across the diagonal.

Excel: Data Analysis → Correlation or multiple `=CORREL(...)` formulas

Correlation matrix: how to compute it with the Data Analysis Toolpak

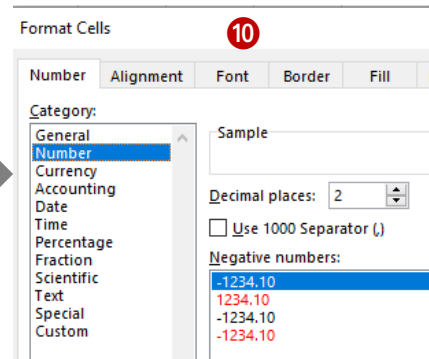


A unique rectangular range,
The correlation function is
quite old and doesn't
accept separate intervals



9 Select the whole matrix

	Income	Limit	Rating	Cards	Age	Balance
Income	1					
Limit	0.707152	1				
Rating	0.703455	0.99688	1			
Cards	-0.01221	0.010231	0.053239	1		
Age	0.123258	0.100888	0.103165	0.042948	1	
Balance	0.432625	0.861697	0.863625	0.086456	0.001835	1



	Income	Limit	Rating	Cards	Age	Balance
Income	1.00					
Limit	0.71	1.00				
Rating	0.70	1.00	1.00			
Cards	-0.01	0.01	0.05	1.00		
Age	0.12	0.10	0.10	0.04	1.00	
Balance	0.43	0.86	0.86	0.09	0.00	1.00

Correlation matrix: how to compute it with the Data Analysis Toolpak

Microsoft Excel



Correlation - Input range must be a contiguous reference.

OK

The typical mistake with the Data Analysis Toolpak

Yes. This is a limitation of the Correlation tool in Excel's Analysis ToolPak. ←

The error message:

"Correlation - Input range must be a contiguous reference."

means that the ToolPak does not accept non-adjacent columns as a single Input Range.

For example, suppose you want to calculate correlations among:

Income, Limit, Rating, Age, Education, Balance

If these columns are separated by other columns in the original Excel Table, you cannot select them using Ctrl + Click and use that selection directly in:

Data → Data Analysis → Correlation

Solution: create a temporary contiguous table

In a new area of the worksheet, or preferably in a new worksheet, copy the numerical variables you want to analyze and place them **next to each other**:

Income	Limit	Rating	Age	Education	Balance
14891	3606	283	34	11	333
106025	6645	483	82	15	903
...	...	...	...	...	...

Ctrl-A within the table (to fully select it), extend selection to the headers too, then copy and paste

Then select the entire contiguous range and go to:

Data → Data Analysis → Correlation

Set:

Grouped By: Columns

Labels in First Row: ✓

Excel will generate a correlation matrix, such as:

	Income	Limit	Rating	Age	Education	Balance
Income	1	...	...	...	...	...
Limit	...	1	...	...	...	...
Rating	...	...	1	...	...	...
Age	...	...	...	1	...	...
Education	...	...	...	...	1	...
Balance	...	...	...	...	...	1

Alternative: use the CORREL function

If you only need the correlation between two variables, there is no need to create a temporary table.

For example:

```
excel
=CORREL(Table1[Age],Table1[Income])
```

Here, the columns do not need to be adjacent. CORREL() simply receives the two arrays whose Pearson correlation coefficient has to be calculated.

So the key distinction is:

Analysis ToolPak → Correlation: requires one contiguous rectangular range.

CORREL() function: can work directly with non-adjacent columns.

Correlation is not causation

A strong relationship can be real and useful without proving that X causes Y.

Common cause

A third variable drives both X and Y.

Example: season affects both heating usage and energy bills.

Reverse direction

Y may influence X rather than X influencing Y.

Correlation alone cannot determine direction.

Spurious relationship

With many variables, some correlations appear by chance.



Does X cause Y?

Not answered by correlation alone



Need additional evidence

Domain knowledge · time ordering · experiments · causal designs.

Correlation in Anti Money Laundering

Suspicious transaction report (STR)

X = Crimes / Bank Branch

Y = STR / Bank Branch

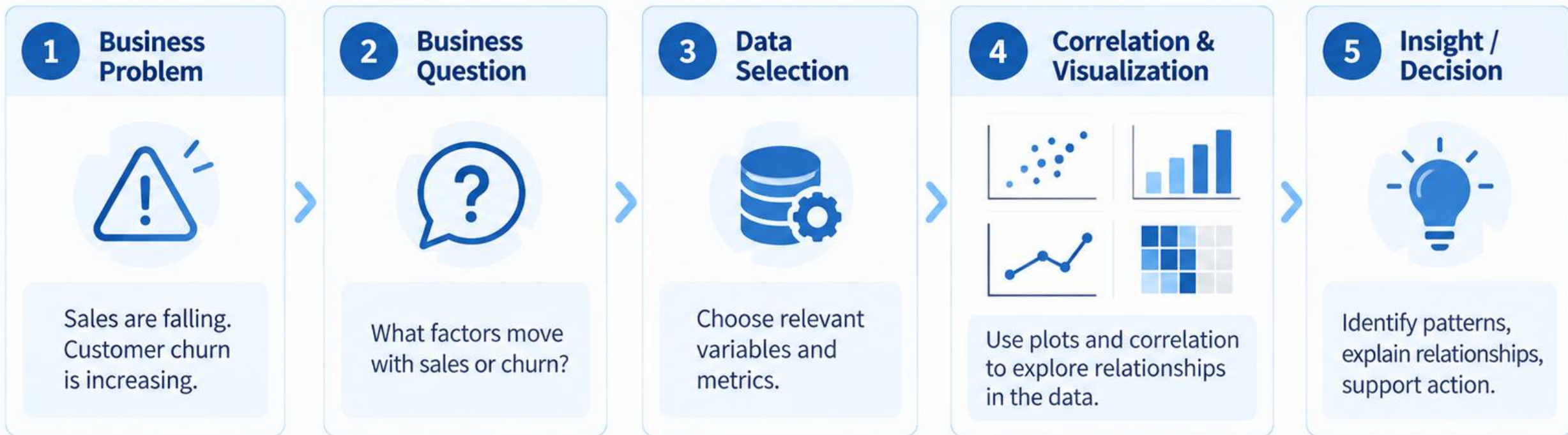
Region	Crimes	STR	Bank Branch	X	Y
Piemonte	100	7100	2364	0.0423	3.0034
Valle d'Aosta	2	212	95	0.0211	2.2316
Liguria	105	2911	822	0.1277	3.5414
Lombardia	204	25373	5805	0.0351	4.3709
Veneto	105	7841	2971	0.0353	2.6392
Trentino - Alto Adige	29	1099	826	0.0351	1.3305
Friuli-V.G. Giulia	19	1488	816	0.0233	1.8235
Emilia-Romagna	85	6979	3009	0.0282	2.3194
Toscana	307	5908	2184	0.1406	2.7051
Umbria	1	949	493	0.002	1.9249
Marche	48	2067	1012	0.0474	2.0425
Lazio	139	9325	2456	0.0566	3.7968
Abruzzo	16	1265	606	0.0264	2.0875
Molise	0	316	128	0	2.4688
Campania	237	9769	1421	0.1668	6.8747
Puglia	153	4519	1252	0.1222	3.6094
Basilicata	11	521	228	0.0482	2.2851
Calabria	40	2127	441	0.0907	4.8231
Sicilia	118	4497	1501	0.0786	2.996
Sardegna	26	1153	597	0.0436	1.9313

Correlation and Data Visualization

Start with a Business Problem

Plots and correlations are tools to answer business questions, not the starting point.

From business questions to better decisions.



**Never start with the chart.
Start with the business need.**

Visualizations and correlations are tools to answer business questions, not the starting point.

Example: Understanding rising defaults



Correlation: typical business reasons in banks

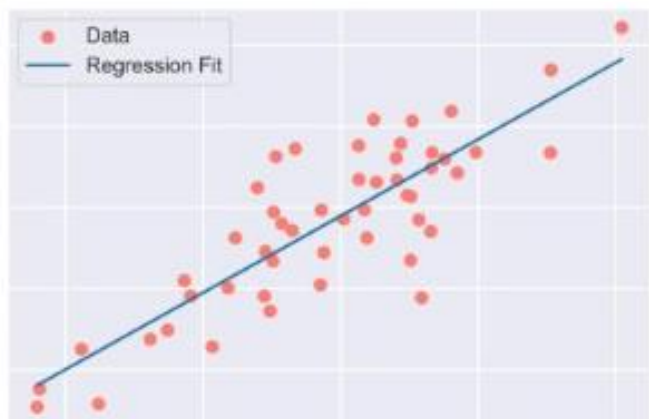
- **Portfolio diversification** — to assess how different assets move together and identify diversification opportunities.
- **Market risk management** — to understand how market variables (e.g. interest rates, FX rates, equity prices) are related and how shocks may propagate.
- **Credit risk assessment** — to measure dependencies between borrowers, sectors, or geographic regions and identify concentration risk.
- **Stress testing** — to estimate how different risk factors may behave together under adverse market scenarios.
- **Liquidity risk management** — to analyse relationships between cash flows, funding sources, deposits, and market conditions.
- **Fraud detection** — to identify unusual relationships between transactions, accounts, merchants, or customer behaviours.
- **Anti-money laundering (AML)** — to detect patterns or dependencies across transactions, accounts, counterparties, and entities that may warrant further investigation.
- **Customer behaviour analysis** — to understand relationships between products and customer behaviours, e.g. credit card usage and loan applications.
- **Interest rate risk management** — to analyse the relationship between interest rates, bond prices, deposits, and lending activity.
- **Forecasting and modelling** — to identify variables that are statistically related and may be useful as inputs to predictive models.
- **Trading and hedging** — to assess whether two instruments tend to move together and therefore may be useful for hedging purposes.
- **Operational risk** — to identify relationships between operational incidents, business units, processes, or external factors.

Correlation can be Misleading



blog.DailyDoseofDS.com

Data Without Outliers



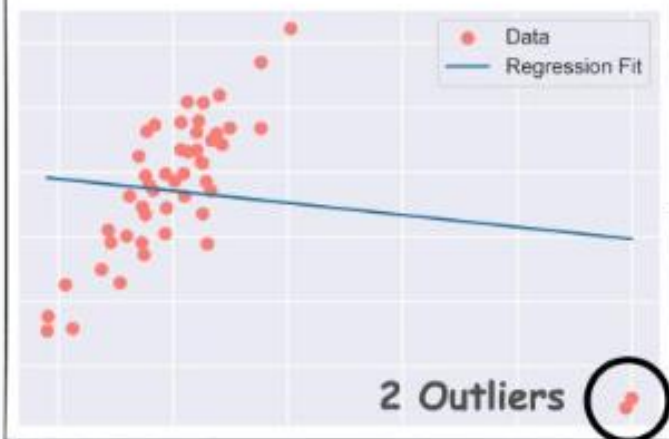
Pearson
Correlation **0.91**

Conclusion



Variables are
positively correlated

Data With Outliers



Pearson
Correlation **-0.05**

Conclusion

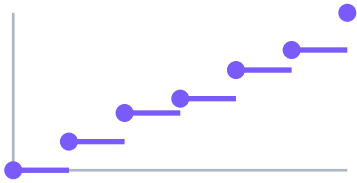


Variables are
almost uncorrelated

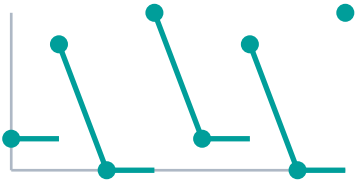
Patterns in data: what EDA is trying to discover

A pattern is a repeatable structure in the data that helps explain, predict or segment observations.

Trend



Seasonality



Clusters



Association



Anomaly



- Patterns are hypotheses until validated — they may reflect sampling, missingness or data quality problems.
- In banking, patterns often differ across segments, products, branches and time periods.
- A useful EDA habit: ask “what would make this pattern disappear?”

Pattern ≠ proof. It is a structured clue that guides the next question.

What is a “pattern”?

Pattern recognition means finding recurring structures inside data that may initially look random.

Look at this set of numbers and letters:

- knsxoi543y77778efh8rfv097777d
- j39e8389f7777lmjbh8ihrz7777
- knsxoi543y77778efh8rfv097777
- 39e8389f7777lmjbh8ihrz>7777

Do you notice anything unusual?

Exactly. After every 10 apparently random letters and numbers, the digit “7” is repeated four times — the pattern “7777”. Your brain has recognised a regular structure inside data that initially looks **random**. This process is called **pattern recognition** (one of a business analyst’s task)

Pattern recognition = identifying recurring models, structures or regularities within raw data.

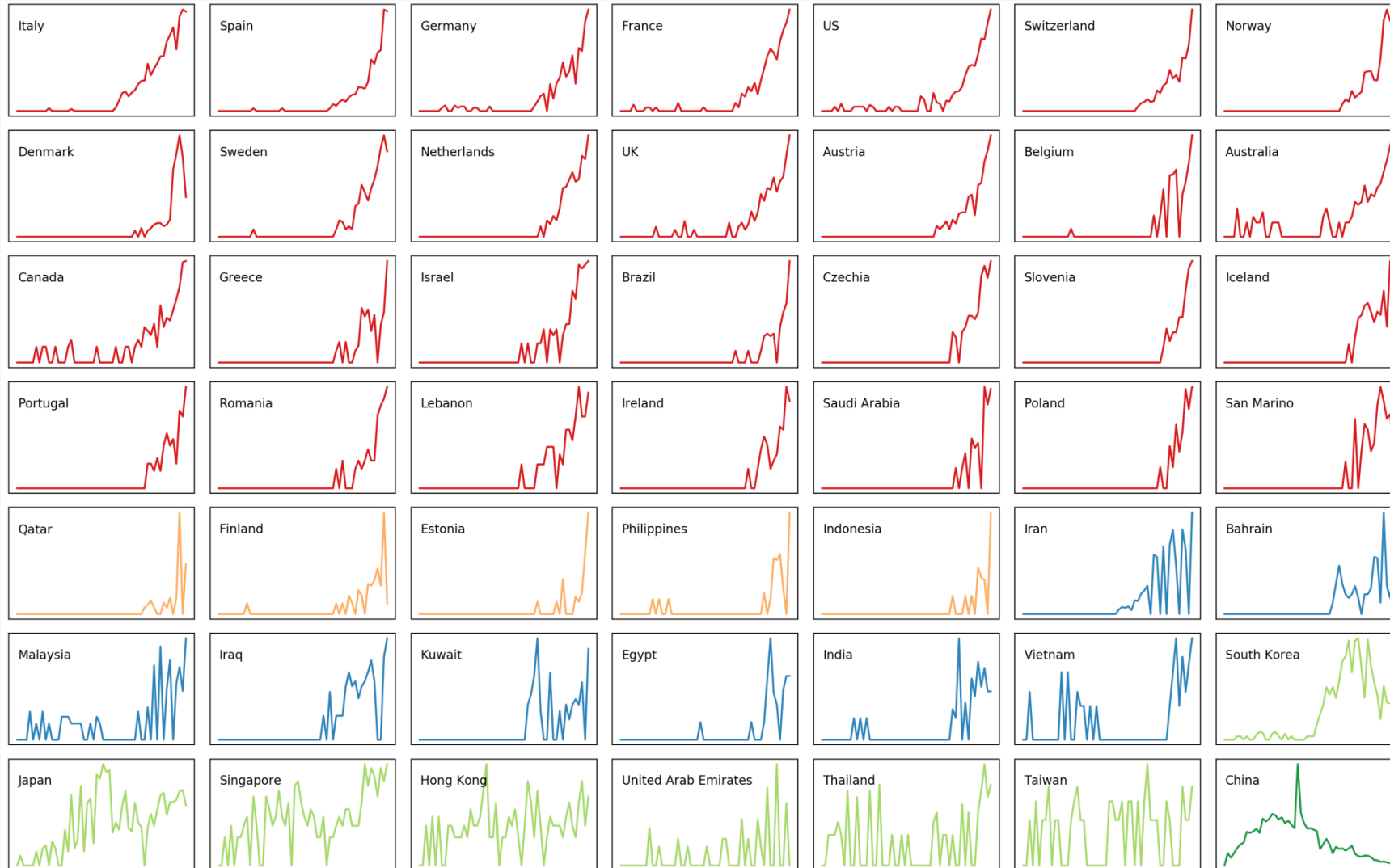
Not yet pre-processed data

Patterns are repeated structures



The same idea applies to data: repeated forms, relationships, sequences or behaviours can be discovered and described.

Patterns



5 different patterns in Covid-19 growth (by colour)

Chapter title



06 – Visualisation

Choose the plot that matches the question

Visualisation is part of reasoning, not decoration.

Distribution of one numeric variable



Histogram / Box plot

Relationship between two numeric variables



Scatter plot

Metric over time



Line chart

Compare categories



Bar / Column chart

Missingness / quality exceptions

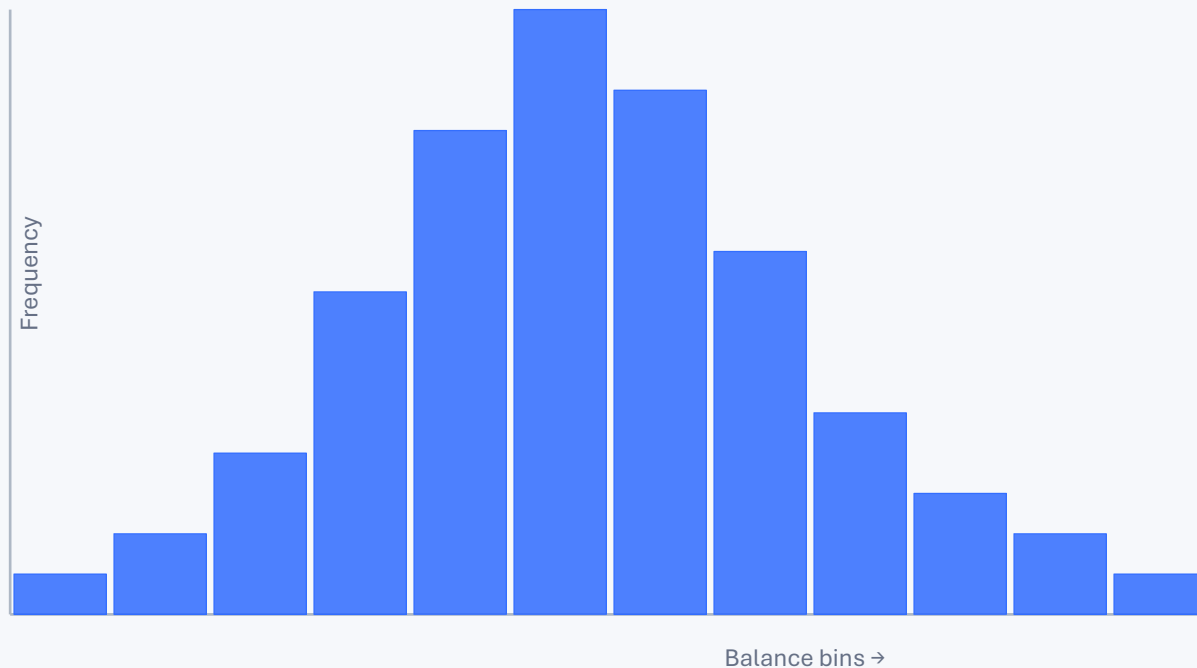


Conditional formatting / table

Avoid charts that hide the data structure. Pie charts, 3D effects and decorative clutter are often poor analytical choices.

Histogram: see the distribution, not only the average

A histogram groups numeric observations into bins and shows how frequently values occur.



Reveals

Skewness · multiple modes · gaps · tails · unusual peaks.

Excel

Insert → Statistical Chart → Histogram, or Data Analysis → Histogram.

Watch the bins

Too few bins hide structure; too many bins create noise.

Question: “What does the distribution of customer balances look like?”

Histogram: why in business?

Two Business Needs for a Histogram

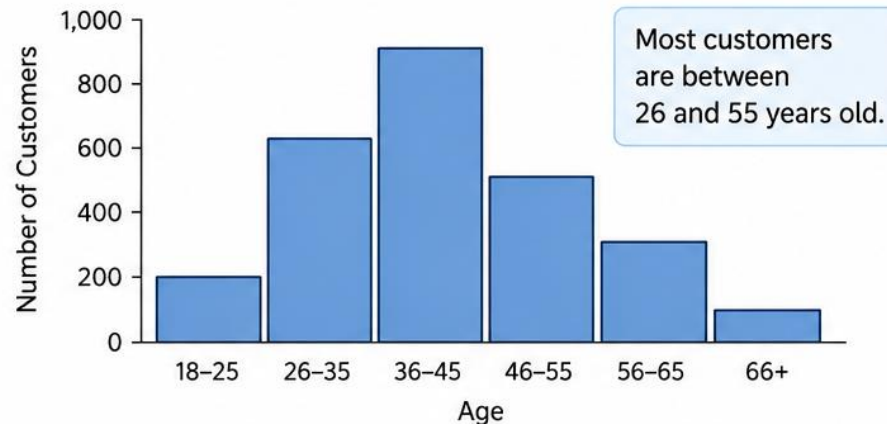
A histogram shows the distribution of a numeric variable, helping you understand the pattern, spread and frequency of values.

1

Understand the Distribution of a Key Metric

A histogram helps you see the overall shape of the data (e.g. typical values, variability, skewness) to support business decisions.

Customer Age Distribution



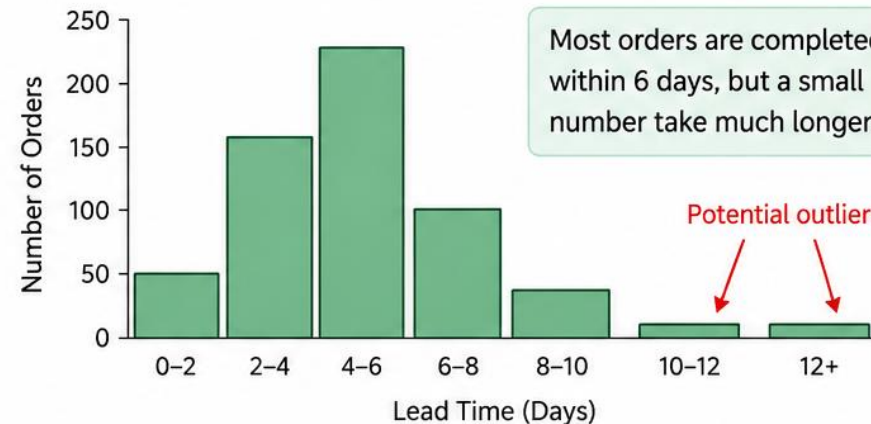
Business outcome: Use this insight to tailor marketing campaigns, product offerings or communication strategies to the main customer segments.

2

Monitor Process Performance and Detect Outliers

A histogram shows how frequently values occur, helping you identify unusual values, variability and whether the process meets the desired range.

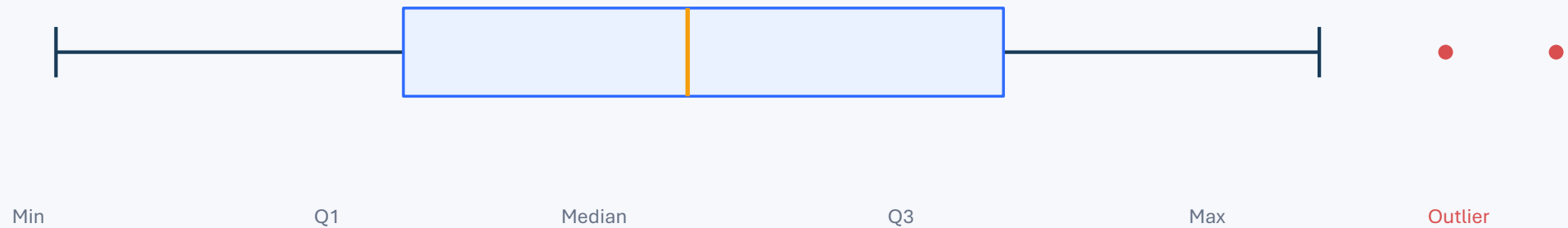
Production Lead Time Distribution



Business outcome: Identify and investigate delayed orders, remove bottlenecks and improve operational efficiency.

Box plot: compact view of centre, spread and outliers

The box spans Q1 to Q3, the line is the median, whiskers show the non-outlier range, and points may mark outliers.



Strength

Very fast comparison across customer segments, products or branches.

Robust centre

Median and quartiles are less sensitive to extreme values than the mean.

Excel

Insert → Statistical Chart → Box & Whisker.

Box plot: why in business?

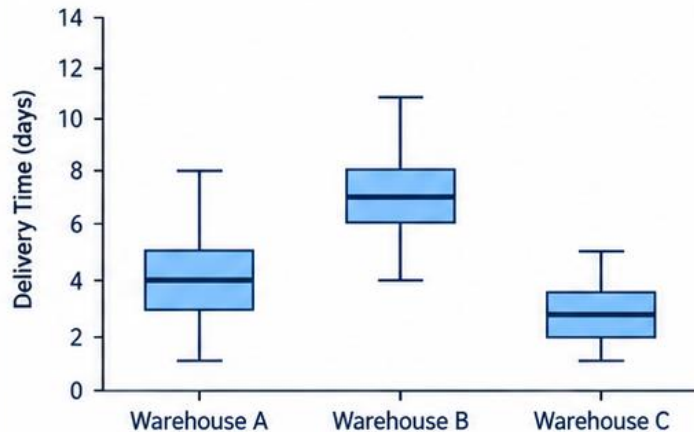
Two Business Reasons to Use a Box Plot

1

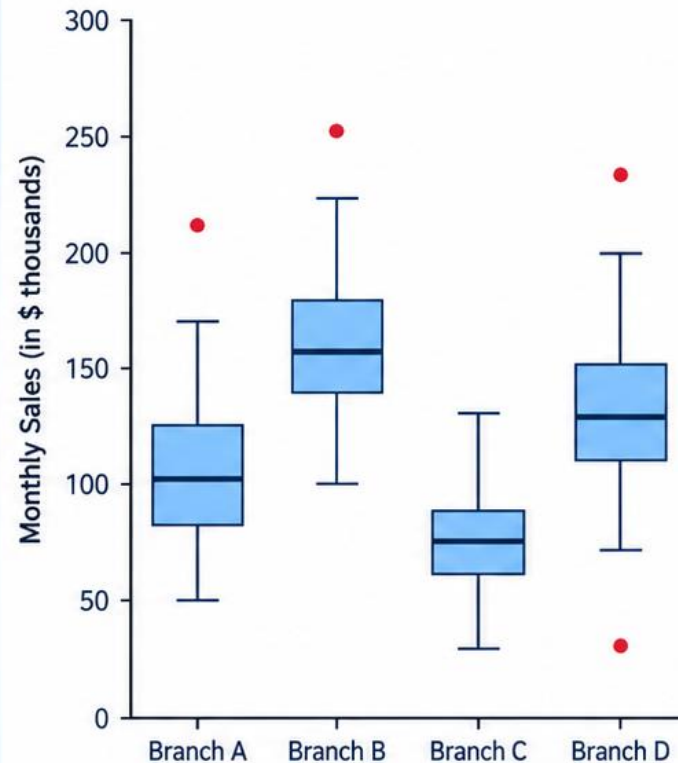
Compare distributions across groups

A box plot helps compare the median, spread, and variability across categories such as branches, product lines, teams, or customer segments.

Example: Delivery Time by Warehouse



Box Plot

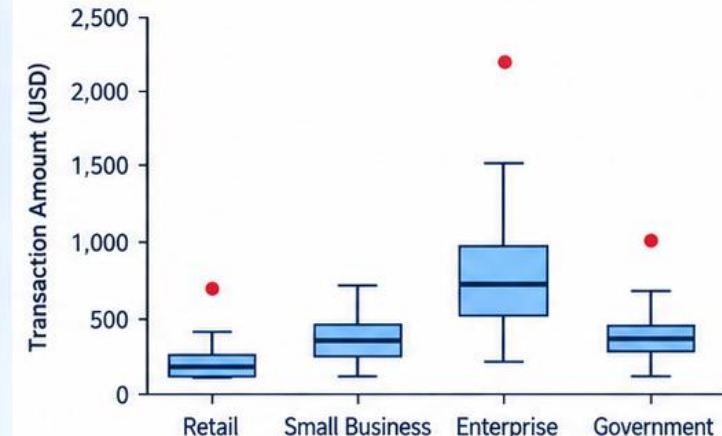


2

Detect outliers and unusual variation

A box plot helps identify unusual values, extreme cases, and groups with unstable performance, for example suspicious transactions, defective batches, unusually slow service times, or abnormal store results.

Example: Transaction Amount by Customer Segment



A box plot summarizes the median, quartiles, spread, and potential outliers.

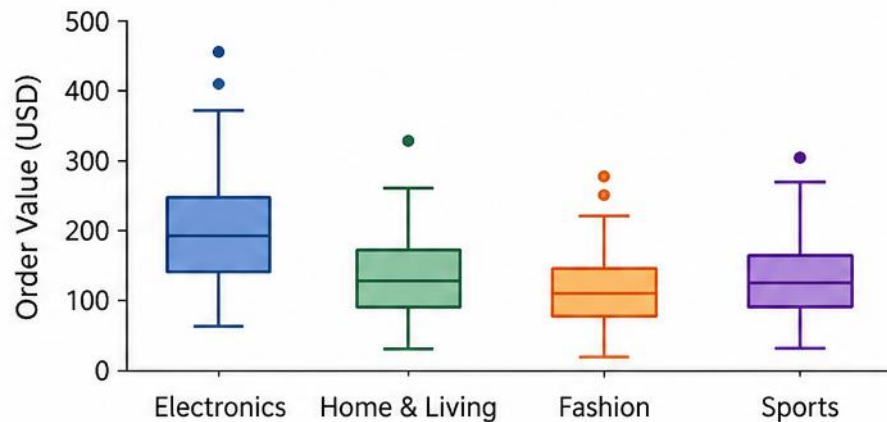
Compared Box plots: why in business?

Two Business Needs for a Comparative Boxplot

1 Compare Performance Across Groups

A comparative boxplot helps identify differences in distribution (median, variability, outliers) between groups, such as product lines, regions or customer segments, to support data-driven decisions.

Order Value by Product Category

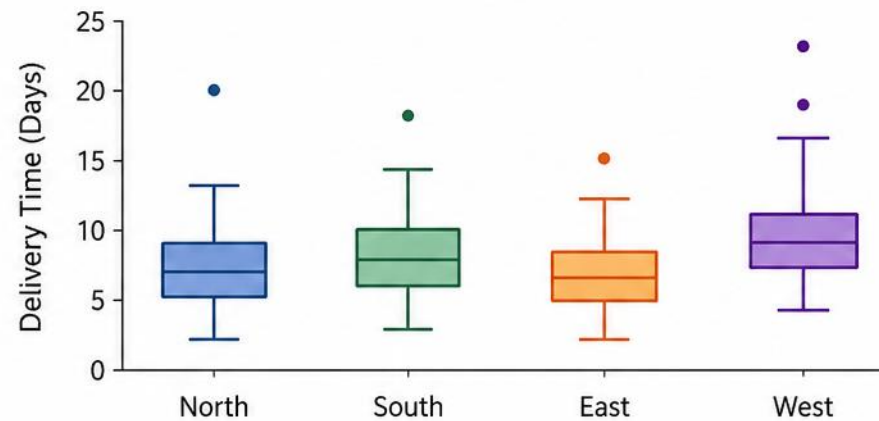


Business outcome: Understand which product categories generate higher order values and which show more variability, to optimize pricing, inventory and marketing strategies.

2 Assess Consistency Across Locations

A comparative boxplot allows you to evaluate and compare the spread and central tendency of a metric across different locations (e.g. stores, branches, or regions), highlighting inconsistencies or outliers that may require attention.

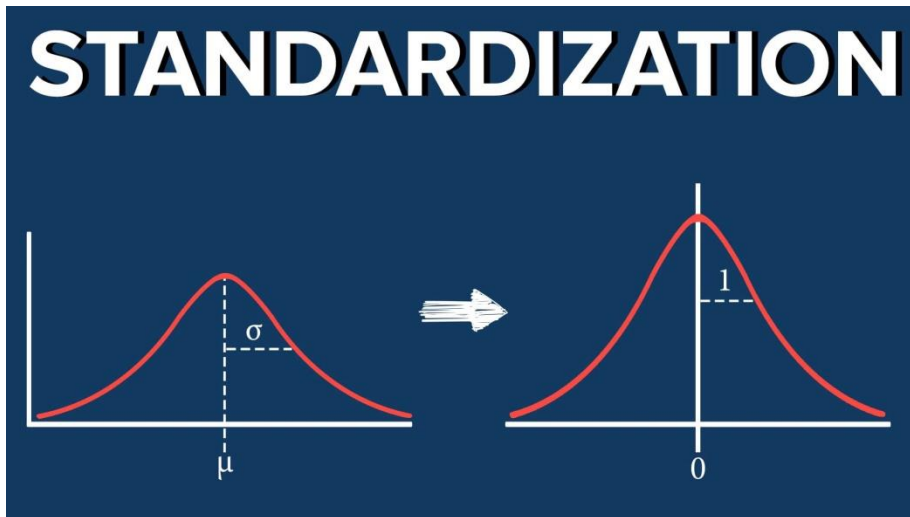
Delivery Time by Region



Business outcome: Identify regions with longer or more variable delivery times to improve operational efficiency and ensure a consistent customer experience across locations.

Box plots: required to compare unevenly distributed data

Two typical useful techniques to compare unevenly distributed data by boxplots → this way, we can then appreciate differences



Normalization Formula

$$X_{normalized} = \frac{(X - X_{minimum})}{(X_{maximum} - X_{minimum})}$$



Standardising a numeric column in Excel (Z-score)

Standardisation expresses each value as a number of standard deviations from the mean.

CONCEPT

Z-score formula

$$z = (x - \mu) / \sigma$$

Interpretation

z = 0 → value is at the mean
z = +1 → one standard deviation above
z = -1 → one standard deviation below

EXCEL FORMULA

For values in A2:A101

```
= (A2 - AVERAGE ($A$2 : $A$101) )  
/STDEV.S ($A$2 : $A$101)
```

Copy the formula down the column.

EXAMPLE

Annual income (k€)

Original	Z-score
20	-1.26
40	-0.63
60	0.00
80	+0.63
100	+1.26

When standardisation is useful

- Variables have very different units or scales
- Distance-based methods are used (e.g. k-NN, k-means)
- You want a common scale centred around zero

Important

Standardisation does not force values into a fixed range. Values below -3 or above +3 are possible and may highlight unusual observations.

Takeaway: standardisation changes the scale, not the relative ordering of the observations.

CONCEPT

$$(x - \min) / (\max - \min)$$

- minimum $\rightarrow 0$
- maximum $\rightarrow 1$
- all other values $\rightarrow$ proportionally between 0 and 1

EXCEL FORMULA

$$= (A2 - \text{MIN}(\$A\$2:\$A\$101)) / (\text{MAX}(\$A\$2:\$A\$101) - \text{MIN}(\$A\$2:\$A\$101))$$

For a 0–100 scale, multiply the result by 100.

EXAMPLE

Original	0–1 scaled
20	0.00
40	0.25
60	0.50
80	0.75
100	1.00

A horizontal number line representing the interval from 0 to 1. The line has tick marks at 0, 0.25, 0.50, 0.75, and 1. The word "minimum" is written below the 0 mark, and the word "maximum" is written below the 1 mark. Four green dots are placed on the line at the positions corresponding to 0, 0.25, 0.50, and 0.75.

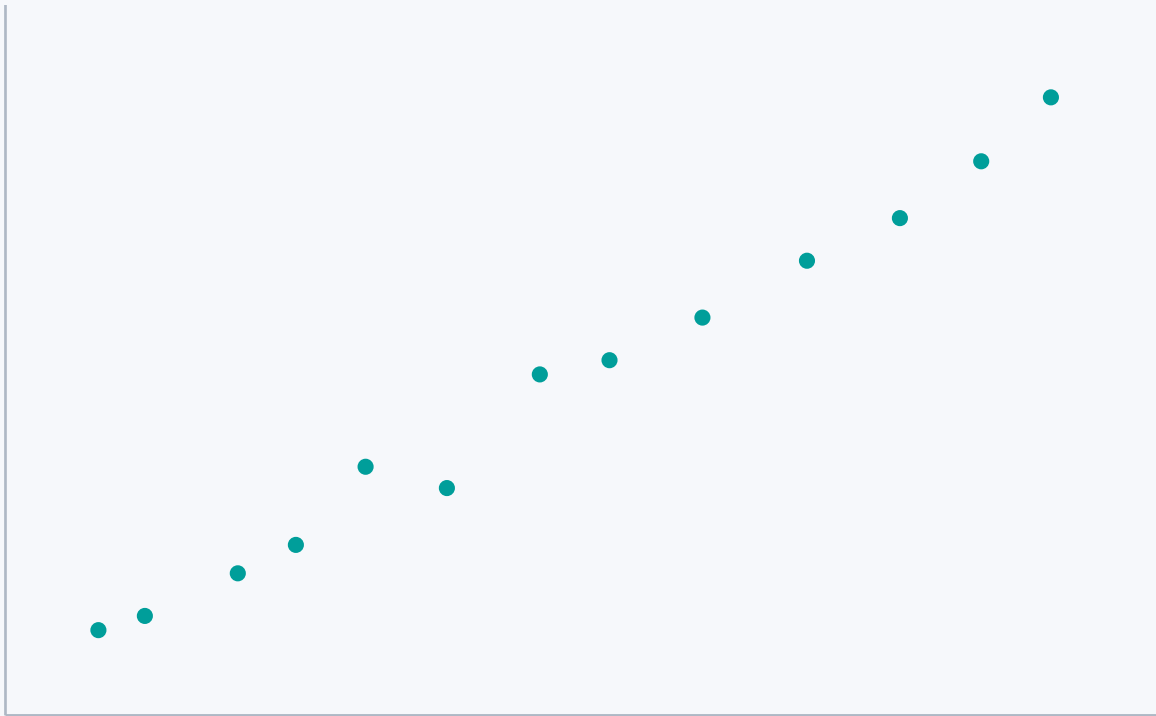
Min-Max normalisation is highly sensitive to extreme minimum or maximum values. One outlier can compress most observations into a very narrow part of the 0–1 interval.

Takeaway: normalisation maps values to a chosen range; standardisation centres and scales them around the mean.

Scatter plot: the essential chart for relationships

Every point is one observation positioned by two numeric variables.

Income vs Balance



Look for

Direction · strength · curvature · groups · isolated points.

Add meaning

Colour points by segment or branch; use labels only for key exceptions.

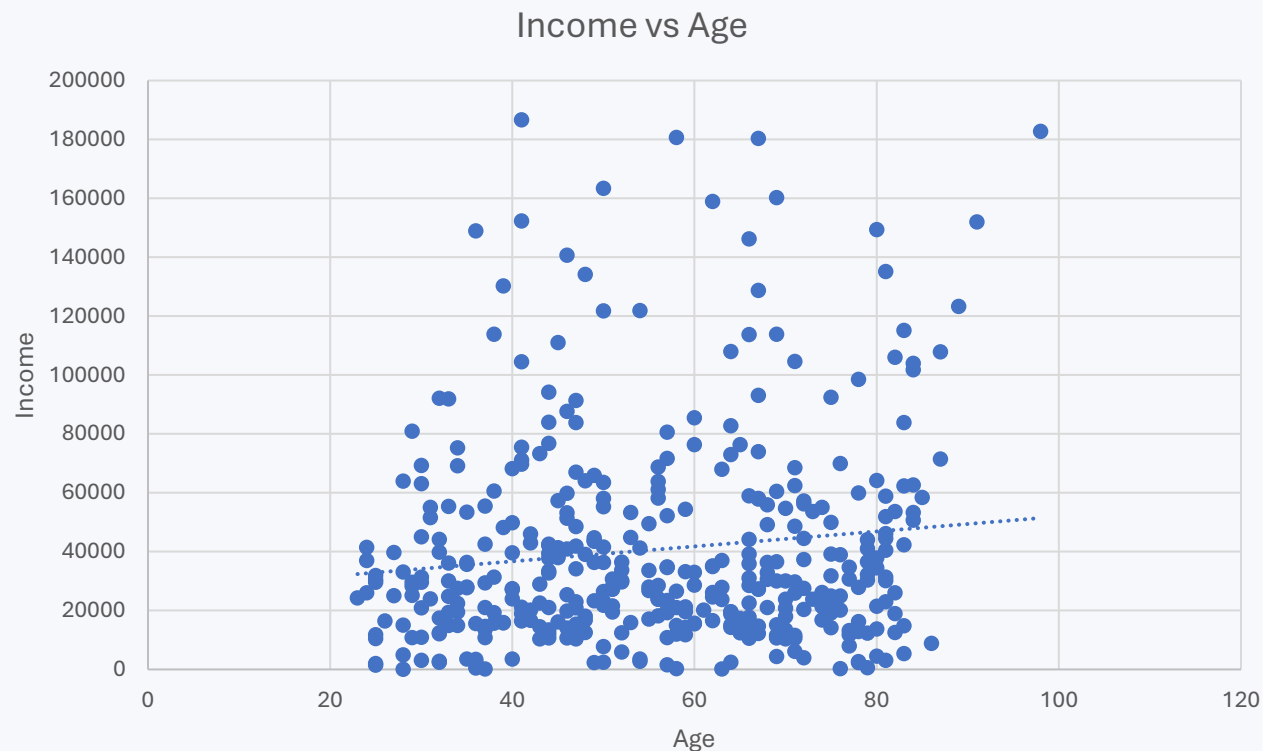
Excel

Insert → Scatter. Add a trendline only when it supports the question.

Pair with `=CORREL(...)` – never interpret the coefficient without seeing the points.

Scatter plot: the essential chart for relationships

The steps of the non-trivial procedure to create a scatter plot



How to Create an Age vs Income Scatter Plot in Excel

1. Select the **Age** column in the Excel Table. **Age will be used as the X variable and therefore displayed on the horizontal (X) axis.** Move the mouse pointer to the top edge of the **Age** header until a **black downward arrow** appears, then click.
2. Go to **Insert → Scatter (X, Y) → Scatter with Only Markers**. Choose the first Scatter chart type: points only, with no connecting lines.
3. Excel will create an initial chart. **Right-click the chart → Select Data.**
4. Remove any series automatically created by Excel by selecting them and clicking **Remove**.
5. Click **Add** and define the new series:
 - **Series name:** **Age vs Income**
 - **Series X values:** select all the values in the **Age** column, excluding the header.
 - **Series Y values:** select all the values in the **Income** column, excluding the header.
6. Confirm by clicking **OK → OK**.
7. Check that the resulting scatter plot has:
 - **Age** on the X-axis
 - **Income** on the Y-axis
 - A single series of points
8. Add the axis titles. Select the chart and go to **Chart Design → Add Chart Element → Axis Titles**. Select both **Primary Horizontal** and **Primary Vertical**.
9. Enter the axis titles:
 - **Horizontal (X) axis:** **Age**
 - **Vertical (Y) axis:** **Income**
10. Set the chart title to **Income vs Age**.

Each point represents **one customer (one row of the Excel Table)**:

(Age, Income)

For example, a customer with **Age = 34** and **Income = 14,891** is represented by the point:

(34, 14891)

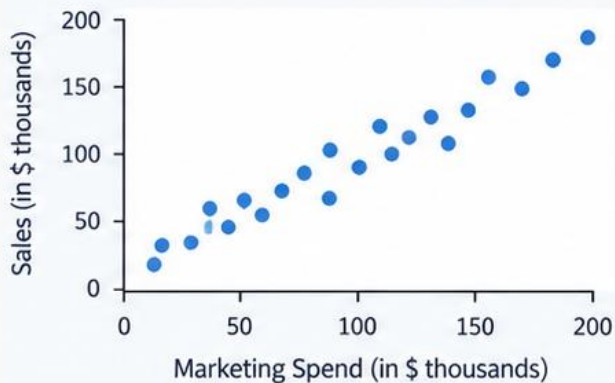
Scatter plot: why in business?

Two Business Reasons to Use a Scatter Plot

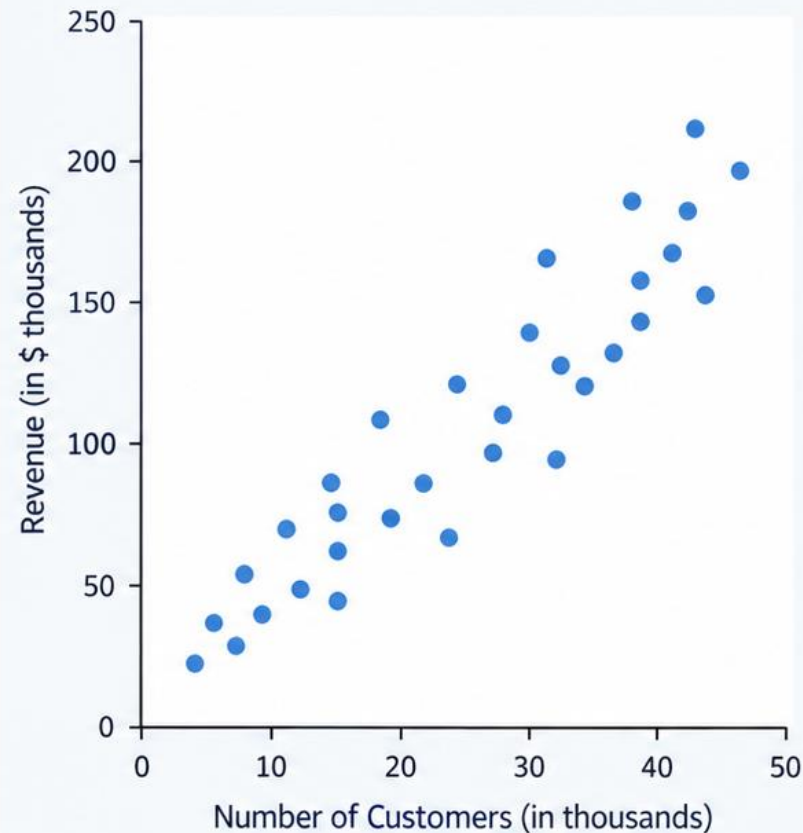
1 Discover relationships between variables

A scatter plot helps reveal whether two variables move together, for example sales vs marketing spend, income vs loan balance, or age vs premium.

Example: Sales vs Marketing Spend



Scatter Plot



2 Detect outliers and unusual cases

A scatter plot helps spot abnormal observations, such as suspicious transactions, unusual customers, underperforming stores, or faulty measurements.

Example: Transaction Amount vs Customer Age



Each dot represents one observation (customer, store, product, month, or transaction).

Line chart: patterns across time

Use line charts when order matters — especially for time series.



Best for

Trend · seasonality · regime shifts · before/after comparisons.

Banking example

Monthly deposits, delinquency rate, call volume or transaction counts.

Caution

Do not connect unordered categories with a line.

Excel: put a real date in the X-axis whenever possible.

Line chart: why in business?



Two Business Needs for a Line Chart

A line chart shows how a metric changes over time, helping you identify trends, patterns and turning points.

1 Track Performance Over Time

A line chart helps you monitor a key business metric over time, understand the trend and assess whether you are meeting your goals.



Business outcome: Evaluate business performance, identify growth (or decline) and make informed decisions (e.g. adjust marketing, sales or resources).

2 Detect Seasonality and Plan Ahead

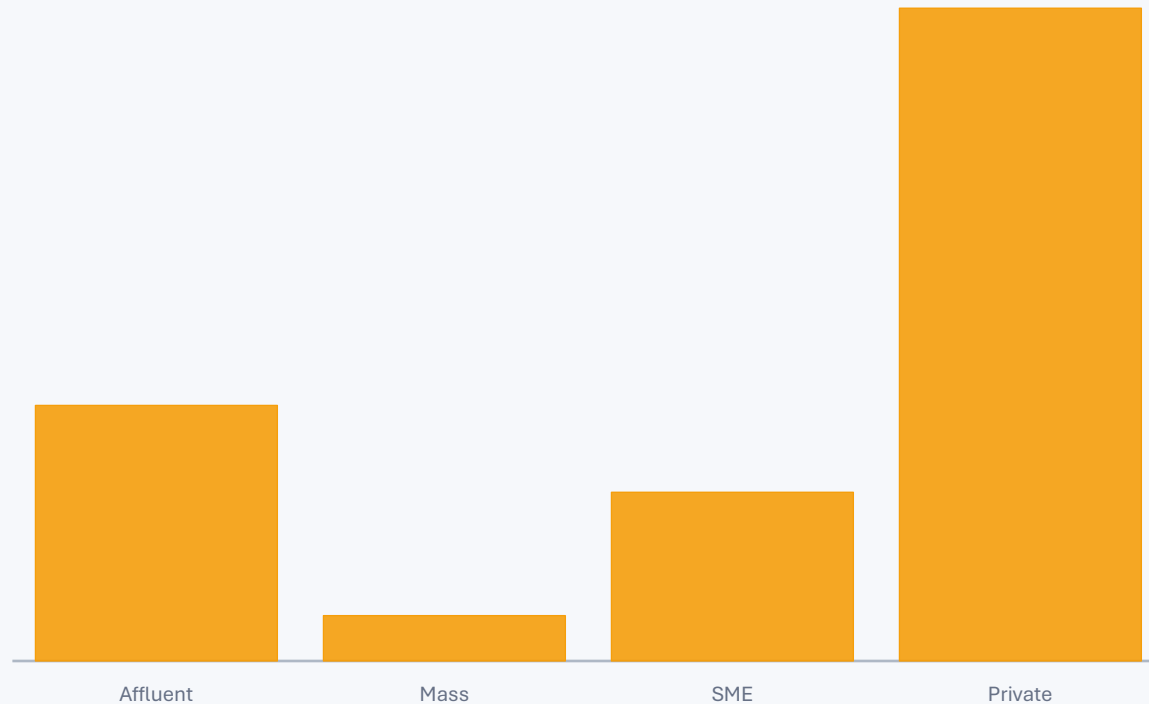
A line chart helps you identify recurring patterns (e.g. seasonal demand) so you can plan inventory, staffing and budget more effectively.



Business outcome: Anticipate seasonal demand, optimize inventory and staffing, and reduce costs while improving customer service.

Bar and column charts: compare categories clearly

The baseline should normally start at zero because bar length encodes magnitude.



Use for

Counts, averages, totals and rates by category.

Good practice

Sort bars when ranking matters; keep category labels readable.

Avoid

3D bars, too many categories and truncated baselines.

Question: “Which customer segment has the highest average balance?”

... to Inform

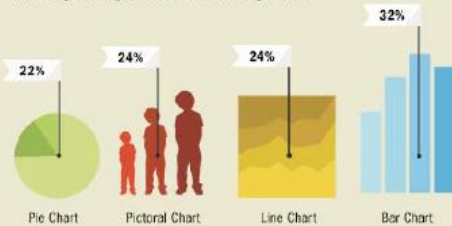
INFOGRAPHIC OF INFOGRAPHICS

Data visualization is a popular new way of sharing research. Here is a look at some of the visual devices, informational elements, and general trends found in the modern day infographic.

DESIGN

CHART STYLE

Percentage of infographics with the following charts:



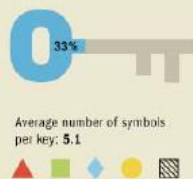
FONT

■ Sans Serif ■ Condensed Sans Serif
■ Serif



KEY INFO

Percentage of infographics with key:



Average number of symbols per key: 5.1

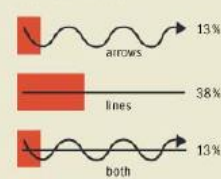


BASE COLOR



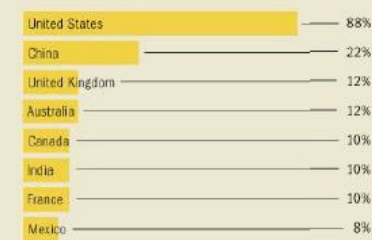
NAVIGATIONAL ICONOGRAPHY

Frequency of arrows & connecting lines in infographics:



CONTENT

COUNTRIES FEATURED



THEME

Relative popularity of different infographic themes:



SECTIONS



CREDITED SOURCES

Average number of sources per infographic: 2.29



TITLE

Average number of words per infographic title: 4.36



CONCEPT & DESIGN Ivan Cash

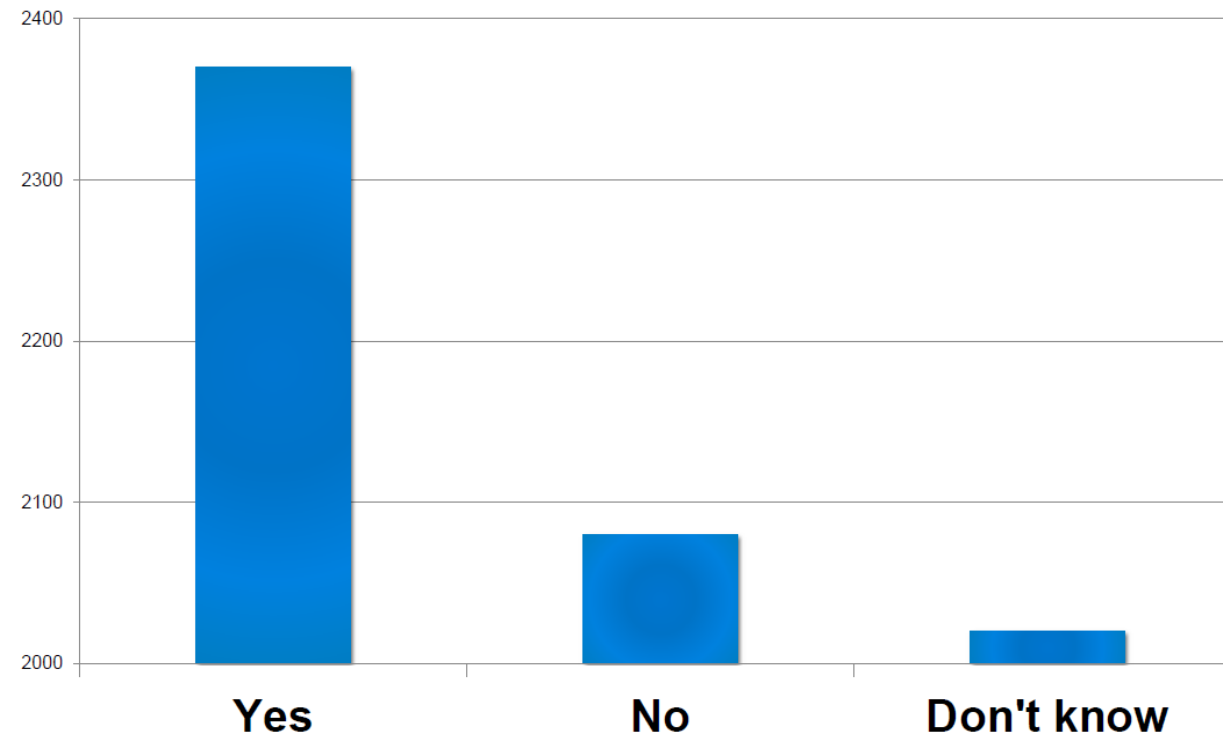
SOURCE 49 infographics collected at random from www.good.is/infographics



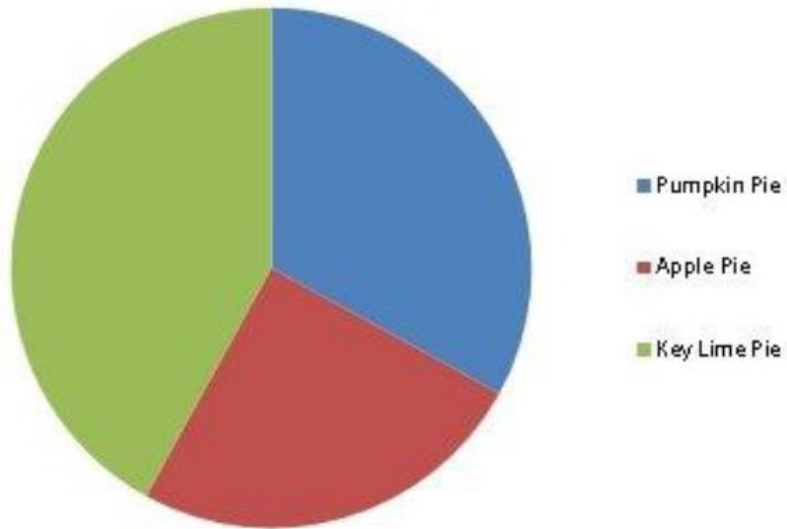
Presentazione di
Microsoft PowerPoint

...to Lie

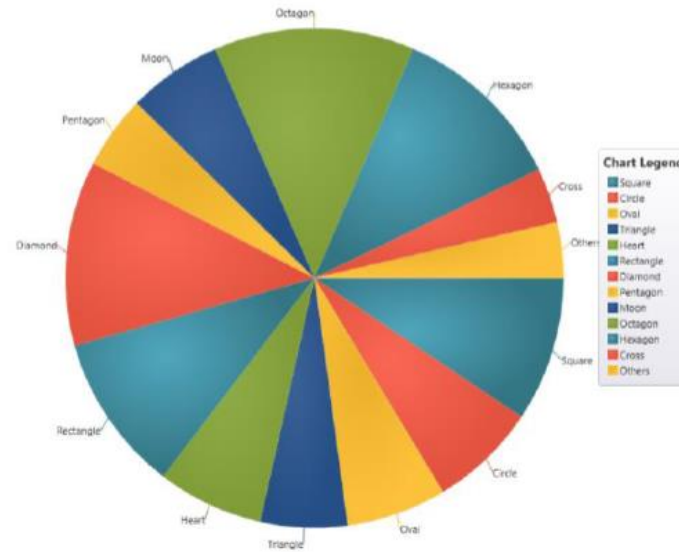
Happy with Management



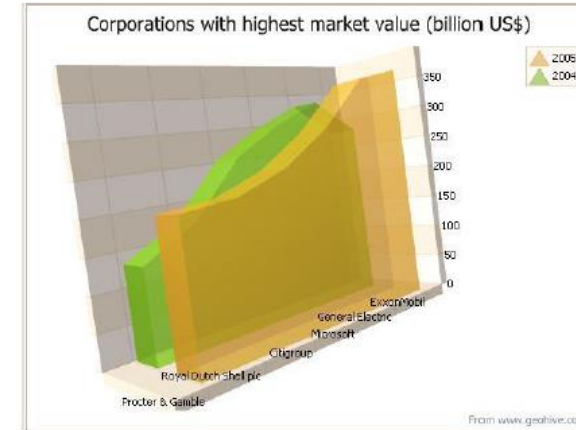
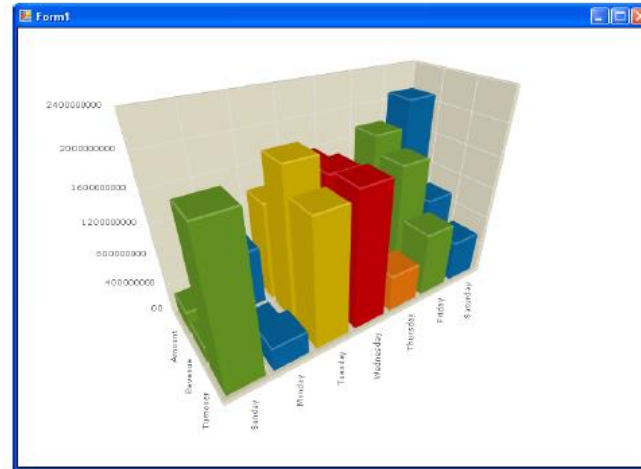
Be Careful with Pie Charts...



Total = 100%
≤ 5 categories
Part to whole
Rough indication



And Never, EVER, use 3D!



Sales Dashboard

(Data as of December 19, 2020)

(All currency is expressed in U.S. dollars.)

Help

The most frequent dashboard

The information:

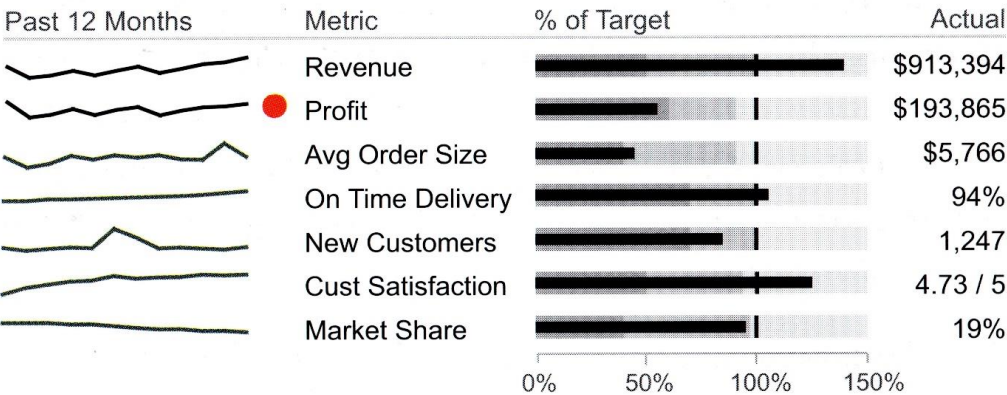
- Sales revenue (*revenue*)
- Sales pipeline *revenue* (i.e., broken down by probability level)
- Profit (*profit*)
- Customer satisfaction
- Top 10 customers
- Market share percentage (*market share*)

For each one:

- Level of aggregation?
- Unit of measure?
- Contextual information?
- Graphical tools?
- Comparison with other measures?
- Positioning?

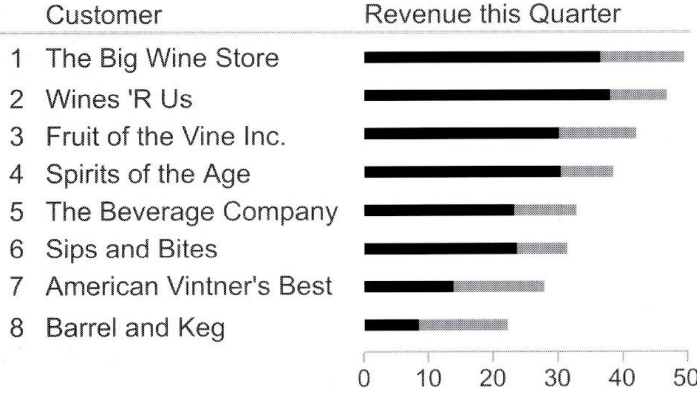
Key Metrics YTD

(■ Actual; | Target; ■ Poor, ■ Satisfactory, ■ Good)



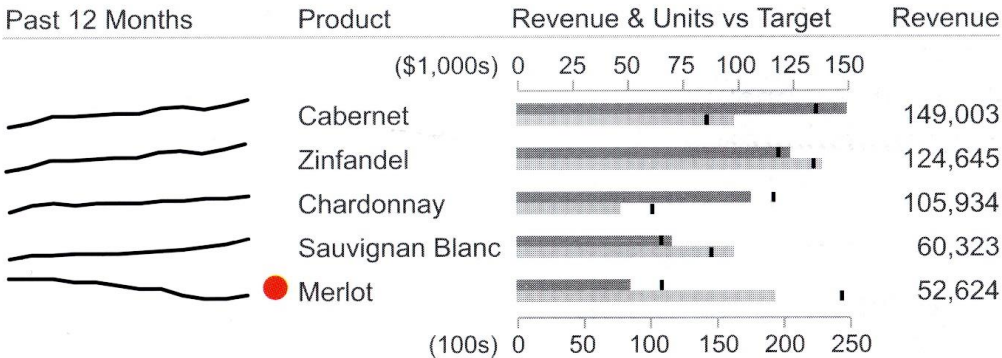
Top 8 Customers This Quarter

(■ Actual; Pipeline: ■)

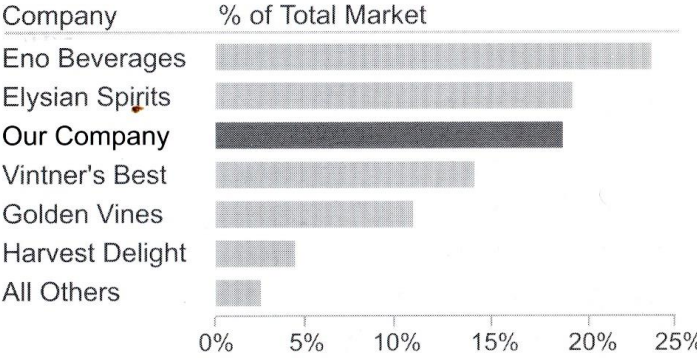


Product Sales YTD

(■ Revenue; ■ Units; ■ Target)



Market Share



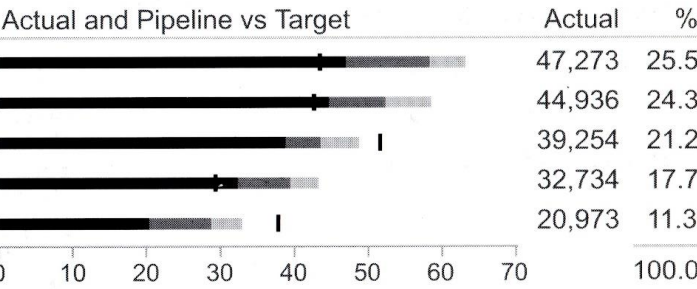
Revenue YTD

(■ Actual; | Target)



Revenue QTD

(■ Actual; Pipeline: ■ 90%, ■ 75%; | Target)



Definition of
year-to-date

What
colours?

What
fonts?

What
plots?

Images?

Sales Dashboard

(Data as of December 19 , 2020)

(All currency is expressed in U.S. dollars.)

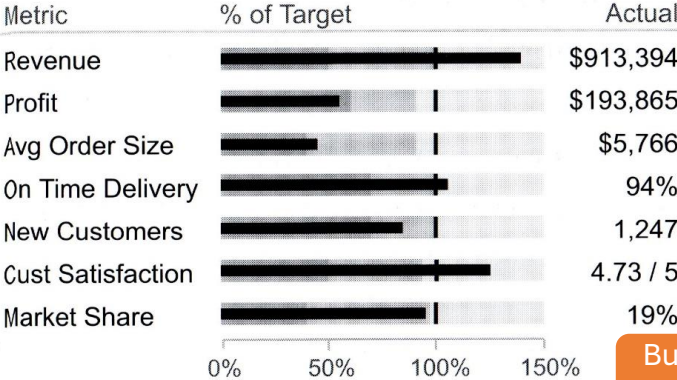
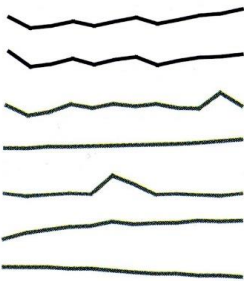
Help

Key Metrics YTD

Upper-left

(■ Actual; | Target; ■ Poor, ■ Satisfactory, ■ Good)

Past 12 Months



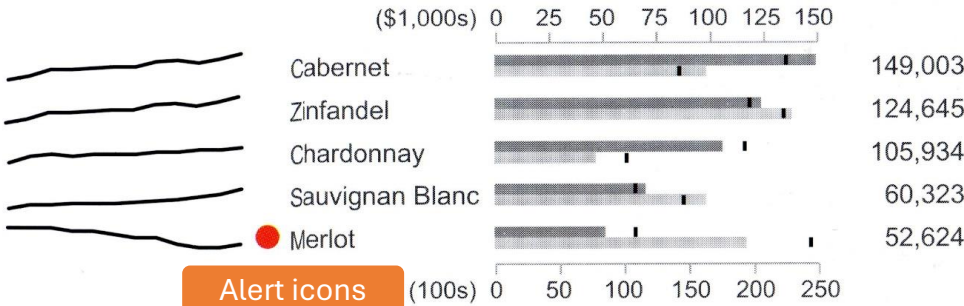
Top 8 Customers This Quarter



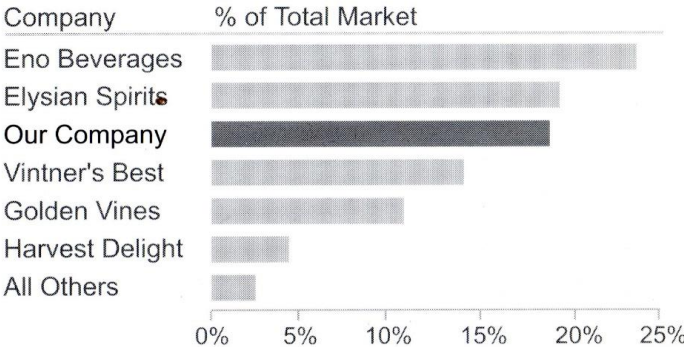
Product Sales YTD

(■ Revenue; ■ Units; ■ Target)

Past 12 Months



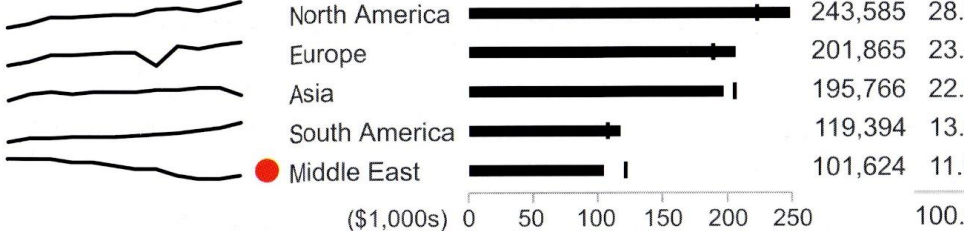
Market Share



Revenue YTD

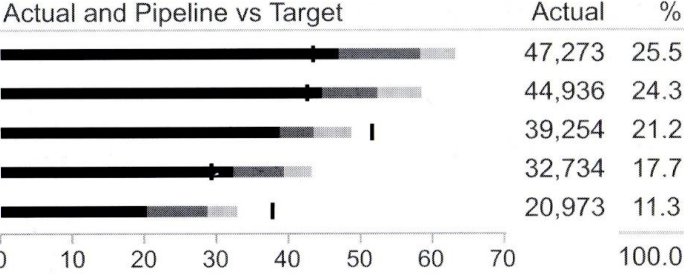
(■ Actual; | Target)

Past 12 Months



Revenue QTD

(■ Actual; Pipeline: ■ 90%, ■ 75%; | Target)



NOTE:

Use colors sparingly

A lot of information in little space

Text and graphics

Stacked bar

Reduction of *non-data pixels*

No contextual help or instructions

Conditional Formatting

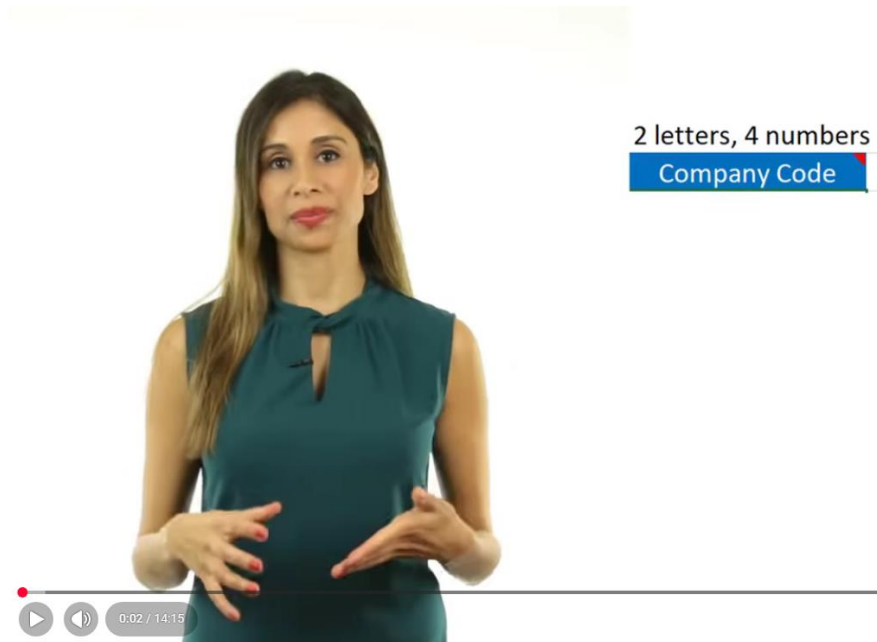
The baseline should normally start at zero because bar length encodes magnitude.



Link to the
video

Data Validation

Strictly speaking, this topic is NOT data visualization. However, while Data validation and conditional formatting in Excel are separate tools, you can easily connect them to create dynamic, interactive spreadsheets.



Excel Custom Data Validation (Use formulas to check for text, numbers & length)
340K visualizzazioni · 8 anni fa

Link to the
video

Chapter title



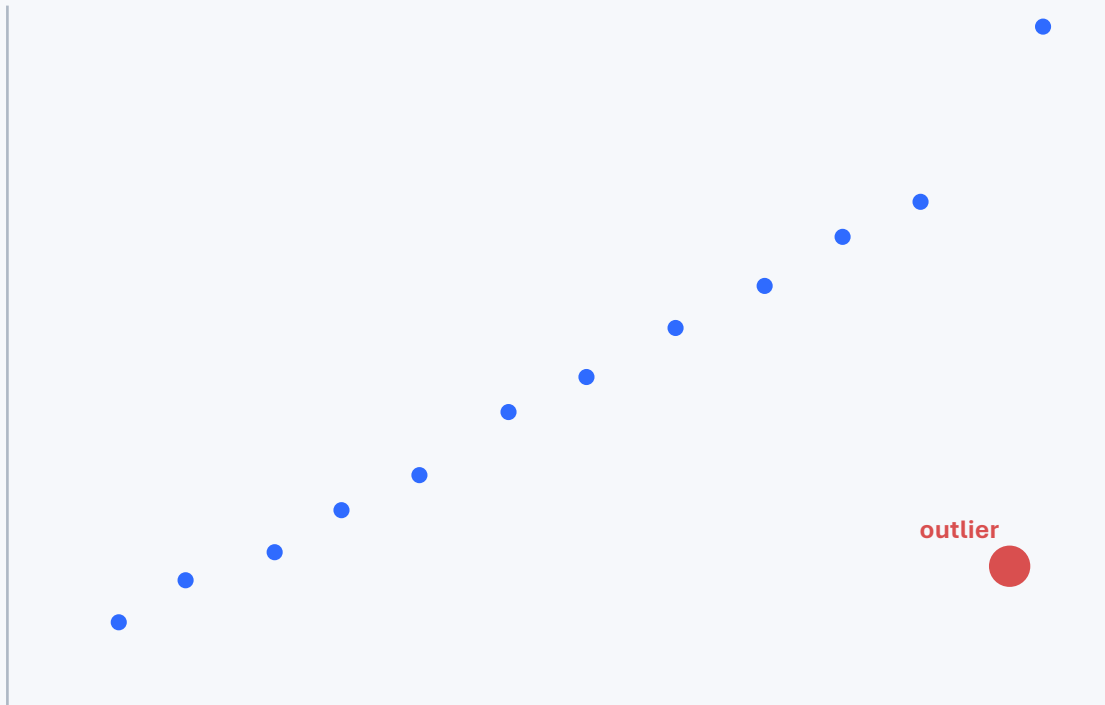
07 – Data issues

Outlier Identification and Management

Outliers: extreme observations that deserve investigation

An outlier may be an error, a rare valid case, a new regime — or the most interesting observation in the dataset.

Mostly regular pattern



Data error

Extra zero, wrong sign, wrong unit, duplicate record.

Valid rare case

Very wealthy customer, large transaction, unusual product.

Process change

System migration, policy change, new branch behaviour.

Interesting signal

Potential fraud, stress event or operational exception.

Never delete an outlier just because it is inconvenient.

Detecting outliers with the IQR rule

The interquartile range is robust and works well as a first-pass rule.



IQR rule

Q1 = 25th percentile

Q3 = 75th percentile

$IQR = Q3 - Q1$

Flag values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.

Q1: `=QUARTILE.INC(range,1)`

Q3: `=QUARTILE.INC(range,3)`

IQR: $=Q3 - Q1$

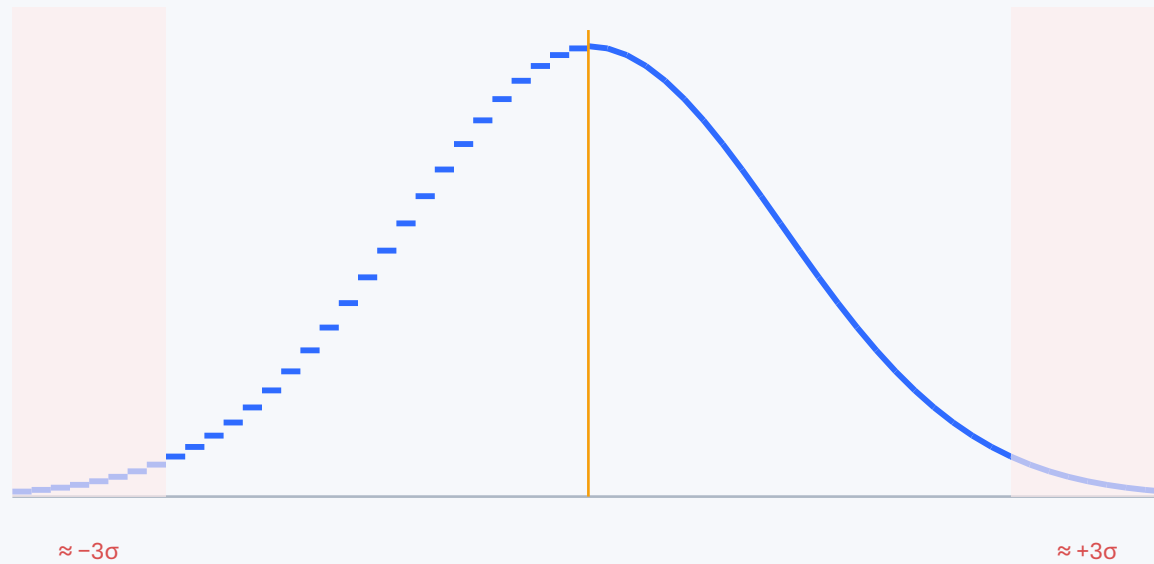
Lower = $Q1 - 1.5 \times IQR$

Upper = $Q3 + 1.5 \times IQR$

A statistical flag is a prompt to investigate, not a business verdict.

Detecting outliers with z-scores

A z-score expresses how many standard deviations an observation lies from the mean.



Formula

$$z = (x - \text{mean}) / \text{standard deviation}$$

```
= (B2-AVERAGE ($B$2:$B$101) ) /STDEV.S ($B$2:$B$101)
```

Caution

z-scores are sensitive to outliers and assume the mean/std are meaningful for the distribution.

For strongly skewed banking data, the IQR rule is often safer as a first screen.

Handling outliers: preserve information and business meaning

The right treatment depends on why the value is extreme.

Validate

Check source system, unit, sign, duplicate and business event.

Correct

Fix confirmed errors using traceable rules.

Keep

Valid extreme cases may be essential to risk analysis.

Transform

Log scale can reduce skew for positive amounts.

Cap / Winsorise

Limit extreme influence when methodologically justified.

Segment / model separately

Different populations may need different treatment.

Document every exclusion or transformation — especially in regulated analytical processes.

Finding Outliers in Excel

Several methods, one goal: identify unusual values that may need investigation

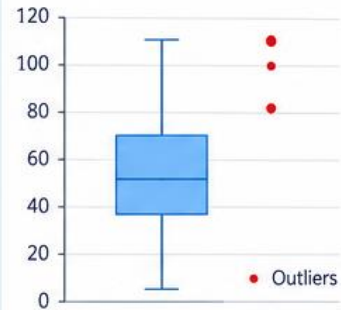
1

Visual inspection

Box plot or scatter plot

Outliers often appear as isolated points in a box plot or scatter plot.

Box plot (example)



In Excel

1. Select the data.
2. Go to Insert → Statistical Chart → Box & Whisker.
3. Look for points outside the whiskers (these are potential outliers).



Available in Excel 2016 and later (Insert → Insert Statistic Chart).

2

Z-score method

Values far from the mean

An observation is an outlier if its z-score is typically greater than 3 (in absolute value).

$$z = \frac{x - \text{mean}}{\text{standard deviation}}$$

Value	Z-score	Outlier?
52	0.12	No
55	0.35	No
58	0.62	No
120	3.48	Yes
49	-0.21	No

In Excel

1. Calculate the mean (AVERAGE) and standard deviation (STDEV.S).
2. Compute the z-score for each value.
3. Flag as outlier if $\text{ABS}(z) > 3$.

Excel formula (in cell C2)

```
= (A2 - AVERAGE($A$2:$A$100)) /  
STDEV.S($A$2:$A$100)
```

3

IQR method

Values outside a typical range

Outliers are values below $Q1 - 1.5 \cdot \text{IQR}$ or above $Q3 + 1.5 \cdot \text{IQR}$.

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower} = Q1 - 1.5 \cdot \text{IQR}$$

$$\text{Upper} = Q3 + 1.5 \cdot \text{IQR}$$

Value	Outlier?
18	No
21	No
22	No
23	No
85	Yes
90	Yes

In Excel

1. Compute Q1 and Q3 using QUARTILE.
2. Calculate IQR, lower and upper bounds.
3. Flag values outside the bounds.

Excel formulas (example range A2:A100)

```
Q1: =QUARTILE.INC(A2:A100,1)  
Q3: =QUARTILE.INC(A2:A100,3)  
IQR: =Q3 - Q1  
Lower: =Q1 - 1.5*IQR  
Upper: =Q3 + 1.5*IQR  
Flag (in B2): =IF(OR(A2<Lower,A2>Upper),  
"Yes","No")
```

4

Modified Z-score

More robust (uses median)

Uses the median and the Median Absolute Deviation (MAD). Consider an outlier if $|M| > 3.5$.

$$M = 0.6745 \cdot \frac{x - \text{median}}{\text{MAD}}$$

Value	Modified Z	Outlier?
18	-0.12	No
21	0.05	No
22	0.18	No
23	0.31	No
85	4.21	Yes
90	5.03	Yes

In Excel

1. Compute the median (MEDIAN).
2. Calculate the MAD (median of absolute deviations).
3. Compute the modified z-score.
4. Flag as outlier if $\text{ABS}(M) > 3.5$.

Excel formula (in B2)

```
=0.6745*(A2-MEDIAN($A$2:$A$100))/  
MEDIAN(ABS($A$2:$A$100)-MEDIAN($A$2:$A$100))
```

5

Domain knowledge

Business rules

Sometimes you know what values are unrealistic based on context (e.g. negative age, very high income).

Value	Rule	Outlier?
-5	Age < 0	Yes
220	Age > 120	Yes
35	—	No
105	—	No

In Excel

1. Define business rules (e.g. min/max allowed values).
2. Use IF statements to flag violations.

Excel formula (example for Age in A2)

```
=IF(OR(A2<0,A2>120),"Yes","No")
```



Tip

Combine multiple methods: use statistical techniques to detect unusual values and domain knowledge to validate them.

Next steps

Identify outliers



Investigate
(e.g. data entry error?)



Decide what to do
(remove, correct, or keep)



Document your decision

Key takeaways

- ✓ There is no single best method.
- ✓ Visualize your data.
- ✓ Use statistical methods and domain knowledge.
- ✓ Always investigate before removing outliers.
- ✓ Document your choices.

Synthetic CSV Dataset with Outliers

1	CustomerID, Age, Income_USD, Monthly_Spend_USD, Debt_USD, City
2	C001, 34, 42000, 1800, 6500, Turin
3	C002, 41, 51000, 2200, 7200, Milan
4	C003, 29, 39000, 1600, 4800, Rome
5	C004, 37, 47000, 2100, 5900, Naples
6	C005, 45, 56000, 2400, 8100, Genoa
7	C006, 32, 44000, 1750, 5300, Bologna
8	C007, 38, 49500, 2050, 6100, Florence
9	C008, 27, 36000, 1500, 4200, Palermo
10	C009, 143, 52000, 2300, 7000, Venice
11	C010, 36, 450000, 2600, 8500, Turin
12	C011, 33, 46000, 12000, 6400, Milan
13	C012, 40, 54000, 2150, -2500, Rome
14	C013, 31, 43000, 1850, 5600, Naples
15	C014, 39, 50500, 2250, 6900, Genoa

Outlier:
Unrealistic age (143)

Outlier:
Extremely high income

Outlier:
Very high monthly spend

Outlier:
Negative debt value

bank_customers_outliers.xlsx (participants): 1,000 customers × 22 columns (demographics, income, balances, transactions, card, mortgage, score, complaints) with filters and a frozen header, plus a **Data_Dictionary** sheet with the plausible range for each column.

bank_customers_outliers_TRAINER.xlsx: same data plus three sheets:

- Statistics** — mean, median, standard deviation, quartiles, IQR fences and z-score thresholds for the 15 numeric columns, all live formulas on the Customers sheet
- Checks** — 13 row-level checks (plausible age, tenure vs age, card spend vs limit, mortgage vs payment, dates, duplicates, score out of range...) with a summary of the counts
- Outlier_Key_Trainer** — the 95 injected anomalies, one per row, with ID, Excel row, columns involved, why it is "to be verified" and how to spot it in Excel

The 8 categories: **A** plausible extreme values (an executive earning €1.85M, a 97-year-old), **B** impossible values (age -3 and 187, a €32M income with extra zeros, an income of "42" in thousands), **C** sentinels and blanks (-999, 9999, empty cells), **D** cross-column inconsistencies (28 years of tenure at age 21, mortgage payment above monthly income, a student earning €240k), **E** future dates and "zero" dates (1900-01-01, 1970-01-01), **F** categorical variants ("lombardia", "Lombardia " with a trailing space, "Male", "Counter"), **G** exact duplicates and near-duplicates with a different ID, **H** a cluster of 12 accounts opened in the same month with 48 transactions of €199 each — normal one by one, anomalous as a group.

A point that lands well in class: on income, the z-score flags 1 outlier only (the €32M inflates the standard deviation and masks everything else), while the IQR rule finds 74. That's the perfect argument for removing sentinels and gross errors first, and only then looking for the "real" outliers.

Dates are formatted as YYYY-MM-DD to stay unambiguous for an international audience; if the participants use US Excel and you'd rather have MM/DD/YYYY, I can switch it.



Watch out for the separator: ; versus , (and comma versus point as the decimal separator) depends on the Windows regional format, not on Office, so if you want the screen to show exactly =IF(A2>1.5, ...) you also need to change the regional format in Settings > Time & Language > Region.

In Windows 11, the path is:

Start → Settings → Time & language → Language & region

Here you can find **Regional format**. The selected format, for example **Italian (Italy)** or **English (United States)**, affects Excel, especially:

- `,` or `.` as the decimal separator
- `;` or `,` as the formula argument separator

For example:

Italian (Italy):

```
</> excel
=IF(A2>1,5; "Yes"; "No")
```

English (United States):

```
</> excel
=IF(A2>1.5, "Yes", "No")
```

If you want to change the **separators specifically**, the most direct path in Windows is:

Control Panel → Clock and Region → Region → Additional settings

There you can change the **Decimal symbol** and the **List separator**.

In Windows 10,

Start → Settings → Time & Language → Region

Then look for **Regional format**.

If you want to change the separators specifically:

Control Panel → Clock and Region → Region → Additional settings

There you can set:

- **Decimal symbol**
- **List separator**

So the wording **"Settings → Time & Language → Region"** is correct for Windows 10.

Missing values: “unknown” is a data state, not a number

Missingness can distort summaries and models — and may itself contain information.

x Missing values			
fx =COUNTBLANK (Table1[Income])			
Customer	Income	Balance	RiskGrade
C1001	65,000	12,500	A
C1002	54,000	8,200	B
C1003		45,500	A
C1004	45,000		C
C1005		165,000	

MCAR

Missing Completely At Random — unrelated to observed or unobserved data.

MAR

Missing At Random conditional on observed information.

MNAR

Missing Not At Random — missingness depends on the missing value or process.

Do not silently replace missing values with zero unless zero is the correct business meaning.

Missing Values Identification and Management

Detecting missing values in Excel

Combine formulas, filters and visual highlighting.

```
=COUNTBLANK (Table1[Income])
```

```
=ISBLANK (C2)
```

```
=COUNTIF (C2:C1000,"")
```

```
=IF (ISBLANK (C2) ,"Missing" ,"OK")
```

Be aware: a formula that returns "" may look blank but is not always the same as a truly empty cell.

x Exception list			
fx =IF (ISBLANK (B4) ,"Missing" ,"OK")			
Customer	Income	Balance	MissingFlag
C1001	65,000	12,500	OK
C1002	54,000	8,200	OK
C1003		45,500	Missing
C1004	45,000		Missing
C1005		165,000	Missing

Handling missing data: choose a strategy, do not hide the problem

Treatment depends on volume, mechanism, business meaning and downstream use.

Leave missing

Best when “unknown” is meaningful and downstream tools can handle it.

Exclude rows

Reasonable only when missingness is limited and exclusion does not bias the sample.

Impute simple values

Median for skewed numeric data; mode for categories — document the rule.

Create missing indicator

A separate flag can capture information in the fact that a value is missing (eg, NA, MV, etc).

Impute with a predictive model

Useful for complex data, but adds assumptions and validation requirements.

Example simple rule: `=IF(ISBLANK(C2), MEDIAN($C$2:$C$1000), C2)`

What are missing values?

Real data is rarely complete. Missing values are common, and it's important to recognise and handle them.

Look at this small dataset:

Customer_ID	Age	Region	Balance	Occupation
C001	25	North	1,200	Engineer
C002		South	850	Teacher
C003	37	NA	2,300	
C004	29	East	N/A	Doctor
C005	NULL	West	670	Artist
C006	45		1,100	Manager
C007	52	Central	?	Retired

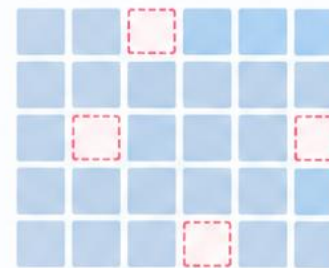
Do you notice anything unusual?



Missing values are data entries that are absent, unknown, not recorded, or not available. They are often represented as blank cells or markers such as NA, N/A, NULL, or ?.

Missing values are pieces of information that should be there, but are not.

Missing values create gaps in our data...



Gaps mean missing information.

They can also appear in real-world forms, such as incomplete records...

Customer Profile
Name:
Age:
Region:
Occupation:
Balance:

Some information is missing here too.

Why they matter



can bias summaries



can distort charts and correlations



can reduce usable sample size



must be handled explicitly

Three types of missing values

How they differ, what they mean, and how to handle them

MCAR

Missing Completely At Random



Definition

The probability that a value is missing is unrelated to the variable itself and unrelated to other observed variables.



Intuition / example

Example: a few survey answers are missing because of a random technical glitch.

ID	Age	Income	Score
1	34	—	88
2	—	52	73
3	28	61	—
4	—	—	90
5	41	48	76

Missing values are scattered randomly across the data.



How to handle

Often the least problematic case. You can use listwise deletion when the amount is small, or simple imputation when appropriate.

Lowest risk of bias



MCAR:

random missingness

MAR

Missing At Random



Definition

The probability that a value is missing depends on other observed variables, but not on the missing value itself once those variables are considered.



Intuition / example

Example: income is more often missing for younger customers; age is observed.

ID	Age	Income	Score
1	22	—	78
2	25	—	85
3	34	56	90
4	47	62	88
5	52	71	76

Missing values in income are more common for younger customers (age is observed).



How to handle

Model-based methods and imputation are usually appropriate: mean within groups, regression imputation, or multiple imputation.

Manageable if related observed variables are available



MAR:

depends on observed data

MNAR

Missing Not At Random



Definition

The probability that a value is missing depends on the unobserved value itself, or on unobserved factors.



Intuition / example

Example: customers with very high debt are less likely to report it.

ID	Debt	Income	Score
1	5	42	80
2	12	58	76
3	28	61	82
4	—	—	74
5	—	—	68

Missing values are concentrated among extreme (typically high) values of the variable itself (e.g., high debt).



How to handle

Most difficult case. Investigate the data-generation process, use domain knowledge, sensitivity analysis, and specialized models when possible.

Highest risk of bias



MNAR:

depends on missing/unobserved data



Key takeaway: the right treatment depends on **why** data is missing, not just on how much is missing.

3 Ways to Handle Missing Values (MV) in Excel

Choose the right method based on your data, the amount of missing values and your analysis goals.

1 Delete rows with missing values (Listwise deletion)

Remove every row that contains at least one missing value.

When to use

- Few missing values
- Missing values are completely random (MCAR)
- You want a simple and clean dataset

	A	B	C	D
	ID	Age	Income	City
1	1	25	50000	Rome
2	2		62000	Milan
3	3	34		Turin
4	4	28	58000	Naples

How to do it in Excel

1. Select the data range.
2. Go to Data → Filter.
3. In each column, filter "(Blanks)" and delete the corresponding rows.



In a dynamic way (newer Excel versions), you can use the **FILTER** function:

```
=FILTER(A2:D100, (A2:A100<>"")*(B2:B100<>"")*(C2:C100<>"")*(D2:D100<>""))
```

2 Impute with the column mean or median

Replace missing values with a simple statistic of the column (mean or median).

When to use

- Small to moderate amount of missing values
- Numeric variables
- You need a quick and simple solution

	A	B	C
	ID	Age	Income
1	1	25	50000
2	2		62000
3	3	34	
4	4	28	58000

How to do it in Excel

Use IF to replace blanks with the mean or median of the column.

Using the mean:

```
=IF(B2="", AVERAGE(B:B), B2)
```

If B2 is empty, replace it with the column mean.

Using the median:

```
=IF(B2="", MEDIAN(B:B), B2)
```

If B2 is empty, replace it with the column median.



Create a new column with the imputed values to keep the original data.

3 Use a predictive model to estimate missing values

Predict missing values using other variables in the dataset (e.g. regression, machine learning).

When to use

- Moderate to large amount of missing values
- Missing values depend on other variables (MAR or MNAR)
- You want more accurate estimates

	A	B	C	D
	ID	Age	Income	Experience
1	1	25	50000	2
2	2		62000	5
3	3	34	54000	8
4	4	28	58000	4

How to do it in Excel (example: linear regression)

1. Go to Data → Data Analysis → Regression.
2. Use the other variables (e.g. Income, Experience) to predict the variable with missing values (e.g. Age).
3. Excel will output the coefficients (intercept and slopes).
4. Use the coefficients in a formula to estimate missing values.

Example formula (after running a regression for Age):

```
=IF(B2="", $E$1+$E$2*C2+$E$3*D2, B2)
```

If B2 is empty, predict Age using Income (C2) and Experience (D2).



This approach usually gives better results than simple imputation, but it requires more work and domain knowledge.

There is no single best method.

Consider the amount of missing values, the reason why they are missing, and the goal of your analysis.

The two files are ready (same 22 predictors as the outlier dataset, new seed, everything in English):

bank_customers_missing.xlsx (participants) — **Customers**: 1,000 customers with 1,598 effectively missing cells (7.3%) under every disguise: empty cells, cells of only spaces, text placeholders (N/A, NA, n.d., ?, -, Unknown), numeric sentinels (-999, 9999, 0, date 1970-01-01), numbers saved as text, 4 ghost records (only the ID), 6 almost-empty records, and structural blanks (no card, no mortgage, dormant account, "NULL = 0 complaints"). Three mechanisms: MCAR, MAR (income missing at 28% for self-employed and business owners vs 3% for the others) and MNAR (the savings balance is missing all the more the higher it is). **Data_Dictionary** documents only the "official" conventions (0 = no card, etc.): everything else is to be discovered.

bank_customers_missing_TRAINER.xlsx — same data plus the workflow sheets, all with live formulas:

- **Codes**: the single list of placeholders, read by all the formulas (add a code and the file updates)
- **Missing_Profile**: per column COUNTBLANK, spaces, placeholders, sentinels, text-numbers, % missing, structural share, what to do + chart
- **Missing_Matrix**: 0/1 map with conditional formatting, count per row, ghost / mostly empty flags
- **Clean**: a single formula pattern per column that normalizes everything to a true blank and converts the text-numbers with VALUE
- **Imputed**: the strategies side by side, always next to the clean value and the missing flag (median, mean, group median with VLOOKUP, derivation with DATEDIF, mode, "Unknown" category, spend/limit ratio, constant 0, listwise on 21 vs 8 columns)
- **Strategies**: reference values, missing rate by occupation, and the trainer section that measures each strategy against **Ground_Truth**
- **Missing_Key_Trainer**: key to everything that was injected, cheat sheet of the strategies, classroom outline with the live numbers

Dates are formatted as YYYY-MM-DD to stay unambiguous for an international audience; if the participants use US Excel and you'd rather have MM/DD/YYYY, I can switch it.

Punti che funzionano in aula (numeri già calcolati nel file): la listwise deletion tiene il 29,5% delle righe (72,3% sulle sole 8 colonne del modello); sul reddito i casi completi danno media 34.752 contro 35.827 vera, la media imputata schiaccia la deviazione standard, la mediana per occupazione recupera 35.514 con metà dell'errore; sul risparmio MNAR le celle mancanti valgono in media 43.669 e noi imputiamo 10.653; la tenure derivata dalla data è esatta 22 volte su 22; la moda del canale indovina il 31%; "NULL = 0 reclami" sbaglia 13 volte su 214.

Allego anche lo script generatore (seed fisso, riproduce entrambi i file). Una nota: nella chat «Dataset outlier per analisi bancaria in Excel» risultava già generata una coppia con questo stesso nome; questa è rifatta da zero, quindi usa i due file insieme — la chiave del trainer vale solo per questo `Customers`.

Data quality: six dimensions to check before trusting analysis

Quality is fitness for use: “good data” depends on the analytical purpose



The same dataset can be “good enough” for one purpose and unacceptable for another.

How data quality changes the analysis

Poor quality can create fake signals, hide real ones and bias decisions.

x Dirty data → distorted evidence		
fx =COUNTBLANK (Table1[Income])		
Issue	Example	Analytical effect
Missing	Income absent for some segments	Biased averages
Wrong type	Amounts imported as text	Rows silently excluded
Duplicate	Customer repeated twice	Inflated counts / balances
Outlier error	Balance ×1000	Mean & correlation distorted
Stale data	Old risk grade	Wrong segmentation



Wrong descriptive statistics

Mean, rate, dispersion and counts become misleading.

Wrong relationships

Spurious or suppressed correlations and segment differences.

Wrong decisions

Mis-prioritised customers, risk, operations or controls.

Data quality is part of the model – not a preprocessing footnote.

Chapter title



08 – Takeaways

Excel EDA checklist: from raw table to trustworthy insight

A compact workflow to use after the course.

- 1 Define the question and the unit of analysis.
- 2 Convert the data to an Excel Table and preserve the raw extract.
- 3 Check types, units, keys and valid categories.
- 4 Profile rows, unique values, ranges and missingness.
- 5 Summarise numeric and categorical variables.
- 6 Visualise distributions, relationships and time patterns.
- 7 Investigate outliers and missing values before changing them.
- 8 Use correlation with scatter plots — not as proof of causality.
- 9 Assess data quality and document every transformation.
- 10 Communicate findings, limitations and next questions.

Good Data Science in Excel = clear question + clean structure + statistical thinking + visual evidence + data quality discipline